

THE DISTRIBUTION OF THE LARGEST NEW EIGENVALUES OF RANDOM COVERS OF GRAPHS

ERIC NASLUND

ABSTRACT. For a connected base graph B , the eigenvalues of a random degree k cover of B split into those of B and $(k - 1)|V(B)|$ new eigenvalues, and the cover is *one-sided Ramanujan* if every new eigenvalue is at most the spectral radius $\rho(B)$ of the universal cover of B . Analyzing 1.4×10^8 samples of the largest new eigenvalue across ten families of random covers, we conjecture that its fluctuations follow the Tracy-Widom law of symmetry class $\beta \in \{1, 2, 4\}$ selected by the Frobenius-Schur type of the associated irreducible representation of the covering group, with centering at $\rho(B)$, an algebraic number for irregular bases, and that the limiting probability that a random cover is Ramanujan is

$$\prod_{[\rho] \neq 1} F_{\beta(\rho)}(0),$$

a product over the non-trivial representations up to conjugation, which the data approaches family by family, to three decimal places where convergence is complete, and where F_β is the Tracy-Widom distribution function in its original normalization. The far tails of every family are localized eigenvalues of polynomially rare defect subgraphs, whose values and rates we compute, prove attainable, and match to the data. The simplest case, covers of a bouquet of loops, is the random d -regular graph, where the limit $F_1(0) = 0.8319081\dots$, not the approximately 52% conjectured by Miller, Novikoff, and Sabelli [59], has since been proven by Huang, McKenzie, and Yau. The agreement of our regular graph data with the proven law validates the pipeline carrying the cover results.

1. INTRODUCTION

A degree k cover (or k -lift) H of a connected base graph B is obtained by assigning to each edge of B a permutation in S_k , and the eigenvalues of H split into the *old* eigenvalues, which are those of B , and the remaining $(k - 1)|V(B)|$ *new* eigenvalues. Random covers were introduced as a combinatorial construction of expanders by Amit and Linial [9, 10] (for expanders see the survey [42]), and the extremal behavior of their new eigenvalues is governed by the spectral radius $\rho(B)$ of the universal cover of B : Friedman [27] formulated the relative Alon conjecture, that the largest new eigenvalue of a random degree k cover of B is at most $\rho(B) + \epsilon$ asymptotically almost surely, successive bounds were given by Linial and Puder [49], by Lubetzky, Sudakov, and Vu [51], and by Puder [64], and the conjecture was proven for regular base graphs by Friedman and Kohler [32, 31] and in general by Bordenave [14] (see also [15, 16]). A sharp quantitative form was given recently by Chen, Garza-Vargas, Tropp, and van Handel [21], and we refer to the survey [71] for the strong convergence framework behind these results. With this in mind, we call a cover H of B *one-sided Ramanujan* if every

Date: July 9, 2026.

2020 Mathematics Subject Classification. Primary 05C80; Secondary 60B20, 05C50.

Key words and phrases. random covers, random lifts, Ramanujan graphs, Tracy-Widom distribution.

new eigenvalue is at most $\rho(B)$. When B is d -regular the universal cover is the d -regular tree and $\rho(B) = 2\sqrt{d-1}$, the threshold of the classical Ramanujan property, and no infinite family does better, by the Alon-Boppana bound (see Nilli [62, 63] and Friedman [26]). When B is the bouquet of $d/2$ loops on a single vertex, a random degree k cover of B is precisely the permutation model of a random d -regular graph on k vertices, with new eigenvalues its non-trivial eigenvalues. Covers also carry the constructive side of the Ramanujan story: Lubotzky, Phillips, and Sarnak [52] and independently Margulis [57] built infinite Ramanujan families when $d-1$ is prime, and whether non-bipartite Ramanujan graphs exist for every degree remains open, though explicit near Ramanujan graphs of every degree and size are known [3]. With covers in hand, Bilu and Linial [13] conjectured that every d -regular graph has a 2-cover whose new eigenvalues all lie in $[-2\sqrt{d-1}, 2\sqrt{d-1}]$, which would settle it. Marcus, Spielman, and Srivastava [56] proved the one-sided statement for bipartite graphs, and Hall, Puder, and Sawin [37] proved that every connected loopless graph has a one-sided Ramanujan cover of every degree. For the group constrained covers studied below, Chandrasekaran and Velingker [20] constructed shift lifts of degrees three and four preserving the Ramanujan property, and Agarwal, Chandrasekaran, Kolla, and Madan [1] bounded the new eigenvalues of random group based lifts, showing in particular that abelian lifts of exponentially large degree are never close to Ramanujan.

The relative Alon theorem localizes the largest new eigenvalue at the threshold but says nothing about which side of it the eigenvalue falls on, and the question of this paper is the finer one: how does the largest new eigenvalue of a random cover fluctuate around $\rho(B)$, and what proportion of random covers is Ramanujan? At the bouquet this is a question with history. Alon [2] conjectured that random d -regular graphs are nearly Ramanujan, Friedman, Kahn, and Szemerédi [30, 25] proved the first $O(\sqrt{d})$ bounds, Friedman [28] proved the conjecture, and Sarnak [65] asked whether a positive proportion is Ramanujan outright, reporting early experiments of Novikoff already pointing to the Tracy-Widom law. The first systematic numerical attack was made in 2008 by Miller, Novikoff, and Sabelli [59], hereafter MNS. Writing $d = \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_V$ for the adjacency eigenvalues of a d -regular graph on V vertices, they sampled 1000 graphs per size from bipartite and perfect matching models with $d \in \{3, 4, 7, 10\}$ and $V \leq 20,000$, with 100,000 graphs at three selected sizes, and, motivated by the numerical evidence of Gaussian Orthogonal Ensemble spacing statistics in the bulk found by Jakobson, Miller, Rivin, and Rudnick [47], gave evidence that the fluctuations of λ_2 follow the $\beta = 1$ Tracy-Widom law of random matrix theory [70]. Writing μ_V and σ_V for the mean and standard deviation of λ_2 over graphs of size V , they fitted power laws

$$2\sqrt{d-1} - \mu_V \sim c_\mu V^m, \quad \sigma_V \sim c_\sigma V^s,$$

found best fit exponents m between -0.72 and -0.83 against s between -0.71 and -0.73 , and were led to conjecture that the limiting probability of being one-sided Ramanujan is the mass of the Tracy-Widom law to the left of its mean,

$$(1.1) \quad F_1(\mu_{TW_1}) = 0.519652\dots,$$

approximately 52%. From a formal expansion of the graph zeta function, Friedman [29] later suggested that the limiting probability exceeds one half, for random regular graphs and for random covers of a regular base alike, without identifying the constant.

We report on a numerical study of the largest new eigenvalue of random covers, roughly a thousand times larger than the experiment of MNS, in a setting in which, outside one result on 2-covers of dense random regular graphs [50], the fluctuation

theory is open. Our data supports a precise and, to our knowledge, new answer: the limiting fluctuations follow the Tracy-Widom law of the random matrix symmetry class dictated by the representation theory of the cover. When the covering permutations are constrained to a group Γ , the new spectrum decomposes into isotypic blocks indexed by the non-trivial irreducible representations of Γ (see section 2), and the Frobenius-Schur indicator of each representation sorts its block into one of the three classical ensembles: real representations give real symmetric blocks (GOE, $\beta = 1$), complex representations give complex Hermitian blocks (GUE, $\beta = 2$), and quaternionic representations give quaternion self-dual blocks (GSE, $\beta = 4$), the threefold way of Dyson [24]. Our main conjecture is that the block fluctuations realize this trichotomy.

Conjecture 1. *Fix $d \geq 3$ and a finite group Γ , let ρ be a non-trivial irreducible representation of Γ with Frobenius-Schur indicator $\varepsilon(\rho) \in \{1, 0, -1\}$, and let A_ρ denote the ρ -isotypic block of a uniformly random Γ -cover of a random simple d -regular base graph on n vertices. Then there is a constant $c_\rho > 0$ such that*

$$\frac{\lambda_{\max}(A_\rho) - 2\sqrt{d-1}}{c_\rho n^{-2/3}} \rightarrow TW_\beta$$

in distribution as $n \rightarrow \infty$, where $\beta = 1, 2, 4$ according as $\varepsilon(\rho) = 1, 0, -1$, and TW_β is the Tracy-Widom law of the corresponding ensemble in its original normalization. In particular the centering is exactly $2\sqrt{d-1}$, with no shift.

The largest new eigenvalue of the cover is the maximum of the block maxima, and blocks of representations that are not complex conjugates of one another behave independently in our data. This yields explicit limiting Ramanujan probabilities.

Conjecture 2. *With the notation of Conjecture 1, the block maxima $\lambda_{\max}(A_\rho)$ are asymptotically independent across non-conjugate non-trivial representations, and the probability that a random Γ -cover is one-sided Ramanujan satisfies*

$$\lim_{n \rightarrow \infty} P\left(\lambda_{\max}(A_\rho) \leq 2\sqrt{d-1} \text{ for all } \rho\right) = \prod_{[\rho] \neq 1} F_{\beta(\rho)}(0),$$

where the product runs over the non-trivial irreducible representations of Γ up to complex conjugation, and F_β is the Tracy-Widom distribution function. Here $F_1(0) = 0.8319081 \dots$, $F_2(0) = 0.96937 \dots$, and $F_4(0) = 0.99857 \dots$.

When all the non-trivial blocks share one symmetry class the largest new eigenvalue itself is Tracy-Widom, and this is what the data shows: $\mathbb{Z}_3$ covers follow TW_2 , quaternion covers follow TW_4 , and permutation covers, whose new spectrum is the single block of the real standard representation of S_k , follow TW_1 . When the blocks mix classes no single Tracy-Widom law can appear, and the data confirms this too: the largest new eigenvalue of a $\mathbb{Z}_4$ cover, a maximum of one GOE block and one GUE block, and of a $\mathbb{Z}_5$ cover, a maximum of two independent GUE blocks, each drift away from every Tracy-Widom law while approaching the predicted maximum law, which the $\mathbb{Z}_5$ family matches to three decimal places (see Table 1 in section 4).

The conjectures have a hyperbolic counterpart. Random degree k covers of a fixed hyperbolic surface are known to have relative spectral gap $\frac{3}{16} - \epsilon$ [53] and in fact $\frac{1}{4} - \epsilon$ [39, 54], the analog of the relative Alon theorem, and for smooth linear spectral statistics of Laplacians twisted by a character of the fundamental group the fluctuations are those of the GOE when the character is real and of the GUE when it is complex [61, 55],

the same selection by Frobenius-Schur type that Conjecture 1 asserts at the edge. No Tracy-Widom law is known or conjectured in that setting.

Our final conjecture extends the picture to covers of a fixed irregular base graph, where no theorem and, to our knowledge, no prior experiment exists. For regular bases, Lubetzky, Sudakov, and Vu [51] simulated random covers of three cubic Ramanujan graphs and conjectured that the probability that a random cover is Ramanujan converges to a constant in $(0, 1)$, and the following conjecture identifies their constant, universally in the base graph.

Conjecture 3. *Let B be a fixed connected base graph of cycle rank $e(B) - v(B) + 1$ at least two, let $\rho(B)$ denote the spectral radius of its universal cover, and let H_k be a uniformly random degree k cover of B . Then there is a constant $c_B > 0$ such that*

$$\frac{\lambda_{\text{new}}(H_k) - \rho(B)}{c_B k^{-2/3}} \rightarrow TW_1$$

in distribution as $k \rightarrow \infty$, where λ_{new} is the largest new eigenvalue. In particular

$$\lim_{k \rightarrow \infty} P(H_k \text{ is one-sided Ramanujan}) = F_1(0) = 0.8319081\dots$$

for every such base graph B .

The $\beta = 1$ law in Conjecture 3 is forced by the picture above, since the standard representation of S_k is real for every k . The hypothesis excludes only the degenerate bases: covers of a tree are disjoint copies of it and covers of a cycle are unions of longer cycles, so neither has a fluctuation theory, while vertices of degree two are harmless, since covers of a subdivision are subdivisions of covers. Taking B to be the bouquet of two loops recovers the permutation model of random 4-regular graphs, where the conjecture is now a theorem (Theorem 4 below), and so Conjecture 3 is the common generalization of the proven regular case to arbitrary base graphs. We test both cases in section 5 at five million covers per size of the fixed bases K_5 and $K_5 - e$: each family selects the TW_1 law, and the empirical Ramanujan probabilities at $V = 10,000$ are 0.83281 and 0.82916 against the conjectured $F_1(0) = 0.8319081\dots$. A third fixed base, $K_4 - e$, tests the conjecture at its boundary, cycle rank exactly two, where the defect structure of the far tail changes and the convergence is slowest. The far tails themselves are not noise: they are localized eigenvalues of polynomially rare defect subgraphs, computable exactly from the universal cover, and in section 5 we prove that every such defect occurs, and produces its eigenvalue, at the polynomial rate governed by its cycle rank (Propositions 5 and 6), conjecture that the defects account for the entire tail, and verify the predicted values and rates in the data.

At the bouquet, the conjectures above meet proven ground. For degrees growing with the number of vertices, Bauerschmidt, Huang, Knowles, and Yau [12] proved Tracy-Widom fluctuations for $V^{2/9+\epsilon} \leq d \leq V^{1/3-\epsilon}$, He [38] for $V^{2/3+\epsilon} \leq d \leq V/2$, and Huang and Yau [45] for the full sparse regime $V^\epsilon \leq d \leq V^{1/3-\epsilon}$, concluding that about 69% of such graphs are Ramanujan and correcting the prediction (1.1) of MNS: the correct one-sided constant is 83%, not 52%. For fixed degree, where Huang and Yau [46] had located λ_2 within $V^{-\epsilon}$ of the threshold and optimal eigenvalue rigidity was then established in [43], Huang, McKenzie, and Yau [44] proved the definitive result.

Theorem 4. *Let $d \geq 3$ be fixed, let G be a uniformly random simple d -regular graph on V vertices, and let $A = d(d-1)/(d-2)^2$. Then*

$$(AV)^{2/3} \left(\frac{\lambda_2(G)}{\sqrt{d-1}} - 2 \right) \rightarrow TW_1$$

in distribution as $V \rightarrow \infty$, where TW_1 is the $\beta = 1$ Tracy-Widom distribution in its original normalization. In particular, since the centering is exactly $2\sqrt{d-1}$,

$$\lim_{V \rightarrow \infty} P\left(\lambda_2(G) \leq 2\sqrt{d-1}\right) = F_1(0) = 0.8319081\dots$$

Unfolding the constants, Theorem 4 states that the mean and standard deviation of λ_2 satisfy

$$(1.2) \quad 2\sqrt{d-1} - \mu_V \approx |\mu_{TW_1}| c(d) V^{-2/3}, \quad \sigma_V \approx \sigma_{TW_1} c(d) V^{-2/3},$$

where $c(d) = \left(\frac{(d-2)^2}{d(d-1)}\right)^{2/3} \sqrt{d-1}$, and where $\mu_{TW_1} = -1.2065335746\dots$ and $\sigma_{TW_1} = 1.2679830577\dots$ are the mean and standard deviation of the un-recentered Tracy-Widom law, so that both of MNS's exponents equal $-2/3$ and the threshold sits $|\mu_{TW_1}|/\sigma_{TW_1} = 0.951537\dots$ standard deviations to the right of the mean. Numerically confirming a proven theorem is not a contribution, and we do not present the regular graph data as one: its role here is to validate the instrument, since the pipeline that recovers the proven law and its constants is the pipeline that carries the cover results.

The conjectures above date to 2018 and were formulated jointly with Lior Silberman, following a suggestion of Peter Sarnak. All of the data reported here was generated by 2023, before the appearance of [44], with every family but the two fixed base ones predating [45] as well, and so the pipeline was in no way tuned to the theorems it now confirms. The value of section 3 then lies in what the theorems do not address, namely the finite size behavior, the diagnosis of MNS's 52%, and the failure of the multigraph model, while the cover families of sections 4 and 5 are the substance of the paper.

We stress three differences between our experiment and that of MNS. First, scale: we use 5×10^6 samples per size against their 1000, sizes up to $V = 500,000$ against their 20,000, and 1.4×10^8 samples in total, so that the four candidate laws, whose distribution functions differ only in the third decimal, are separated by dozens of binomial standard errors rather than fractions of one. Second, methodology: instead of fitting a single power law across all sizes we track the local, consecutive size exponents of the mean gap and standard deviation, which reveals that a global fit over $V \leq 20,000$ systematically overestimates the mean exponent, the precise finite size artifact behind (1.1). Third, and principally, scope: the nearest precursor to our cover experiments is the simulation of [51], one thousand covers per size of three cubic Ramanujan bases, which observed a stable Ramanujan proportion without identifying a law. The representation theoretic selection of the symmetry class, the Tracy-Widom identification, and the explicit constants have, to our knowledge, no analog in any published experiment. Throughout, we study the one-sided question only, and make no claims about the joint behavior of the smallest new eigenvalue.

1.1. Outline. In section 2 we describe the random cover model, the isotypic decomposition, the statistics we compute, and the dataset. In section 3 we present the regular graph control experiment: the convergence to the proven law (1.2), the local exponent diagnosis of MNS's conjecture, and the failure of the permutation multigraph model. Section 4 presents the evidence for Conjectures 1 and 2 across abelian and quaternion covers, together with a quantitative account of the mixed class families. Section 5 presents Conjecture 3, the data for the three fixed base families, and the defect analysis of the far tails, including the two propositions and the completeness conjecture that organize them. Section 6 collects open problems.

2. THE NUMERICAL MODEL AND THE DATASET

2.1. Random Covers and the Isotypic Decomposition. Let B be a connected base graph on n vertices, possibly with loops and multiple edges. A degree k cover of B is determined by a voltage assignment σ giving each oriented edge $e = (u, v)$ of B a permutation $\sigma_e \in S_k$, with $\sigma_{\bar{e}} = \sigma_e^{-1}$ on the reversed edge: the cover H has vertex set $V(B) \times \{1, \dots, k\}$, with (u, i) adjacent to $(v, \sigma_e(i))$ for each edge $e = (u, v)$. A *random cover* of degree k assigns independent uniform permutations to the edges of B . More generally, given a finite group Γ of order k , a random Γ -cover assigns independent uniform elements of Γ to the edges, acting on the fiber Γ by right multiplication. The eigenvalues of B lift to H , and are called the old eigenvalues, and the remaining $n(k-1)$ eigenvalues are called *new*, and we write $\lambda_{\text{new}}(H)$ for the largest new eigenvalue.

The spectrum of a Γ -cover organizes along the irreducible representations of Γ . Decomposing the permutation action of Γ on its group algebra into irreducibles, the adjacency matrix of H is conjugate to a direct sum

$$(2.1) \quad A_H \cong \bigoplus_{\rho \in \text{Irr}(\Gamma)} A_{\rho}^{\oplus d_{\rho}}, \quad A_{\rho} = \sum_{e=(u,v)} E_{uv} \otimes \rho(\sigma_e),$$

where the sum runs over the oriented edges of B , E_{uv} is the matrix unit, and d_{ρ} is the dimension of ρ , so that A_{ρ} is an nd_{ρ} dimensional Hermitian matrix (see [37, Section 2] for this decomposition, and [1] for its use on group based lifts). The trivial representation recovers the base, $A_1 = A_B$, and so the new spectrum of H is the union of the spectra of the non-trivial blocks A_{ρ} , each with multiplicity d_{ρ} . Note that the blocks of complex conjugate representations satisfy $A_{\bar{\rho}} = \overline{A_{\rho}}$ and have identical spectra, and so the distinct sources of randomness are indexed by the non-trivial representations up to conjugation. For a full symmetric cover the same decomposition applies with $\Gamma = S_k$ acting on $\{1, \dots, k\}$: the permutation representation splits as the trivial plus the standard $(k-1)$ dimensional representation, and the new spectrum is the single block of the standard representation.

The Frobenius-Schur indicator

$$\varepsilon(\rho) = \frac{1}{|\Gamma|} \sum_{g \in \Gamma} \chi_{\rho}(g^2) \in \{1, 0, -1\}$$

sorts the irreducible representations into real ($\varepsilon = 1$), complex ($\varepsilon = 0$), and quaternionic ($\varepsilon = -1$) types, and correspondingly sorts the blocks A_{ρ} into the three classical matrix symmetry classes: a real representation may be realized over $\mathbb{R}$, making A_{ρ} conjugate to a real symmetric matrix, a complex representation gives a complex Hermitian block with no further symmetry, and a quaternionic representation gives a self-dual block whose eigenvalues carry an even multiplicity. In random matrix theory these are the symmetry classes of the GOE, GUE, and GSE, whose largest eigenvalues fluctuate according to the Tracy-Widom laws TW_1 , TW_2 , and TW_4 [69, 70]. This threefold way [24] is the mechanism behind Conjecture 1. Our families realize all three types: for $\Gamma = \mathbb{Z}_k$ the non-trivial characters $\chi_j(x) = e^{2\pi i j x / k}$ give n dimensional blocks with entries $\chi_j(\sigma_e)$ on the edges of the base, complex Hermitian in general and real symmetric exactly when $\chi_j = \pm 1$ is real valued, while the quaternion group Q_8 has a unique non-trivial higher dimensional representation, the two dimensional representation of quaternionic type, whose real form is four dimensional.

Concretely, $\mathbb{Z}_3$ has one conjugate pair of complex characters, and so the largest new eigenvalue of a random $\mathbb{Z}_3$ cover is the largest eigenvalue of a single GUE class block.

$\mathbb{Z}_4$ has the real character $\chi_2 = (-1)^x$ together with one conjugate pair, and so its largest new eigenvalue is the maximum of a GOE class and a GUE class maximum. $\mathbb{Z}_5$ has two conjugate pairs, giving the maximum of two GUE class maxima with identical marginal distributions. In general, the non-trivial characters of $\mathbb{Z}_k$ form $\lfloor (k-1)/2 \rfloor$ conjugate pairs, together with the real character $\chi_{k/2}(x) = (-1)^x$ when k is even, and so the largest new eigenvalue of a random cyclic cover is the maximum of $\lfloor (k-1)/2 \rfloor$ independent GUE class maxima and, for even k , one GOE class maximum, with limiting Ramanujan probability

$$F_2(0)^{\frac{k-1}{2}} \quad \text{for odd } k, \quad F_1(0) F_2(0)^{\frac{k-2}{2}} \quad \text{for even } k$$

under Conjecture 2. The quaternion family samples the four dimensional real form of the quaternionic block of a random Q_8 -cover directly, a $4n$ dimensional matrix all of whose eigenvalues are new, with each eigenvalue doubled by the quaternionic structure. These five families, together with the permutation covers of the bouquet (random 4-regular graphs), of K_5 , of K_5 minus an edge, and of K_4 minus an edge, are the subject of sections 3 through 5.

2.2. The Statistics and the Tracy-Widom Constants. By TW_β we mean the Tracy-Widom law in the original normalization of Tracy and Widom [69, 70], with distribution function F_β , mean μ_{TW_β} , and standard deviation σ_{TW_β} , and by $\widetilde{TW}_\beta$ the standardization to mean 0 and variance 1. The constants we use are

	β	μ_{TW_β}	σ_{TW_β}	$ \mu_{TW_\beta} /\sigma_{TW_\beta}$	$F_\beta(\mu_{TW_\beta})$
(2.2)	1	-1.2065335746	1.2679830577	0.9515376	0.519652
	2	-1.7710868074	0.9017731382	1.9640048	0.515016
	4	-2.3068848932	0.7195302084	3.2060987	0.511069

following [18, 17], and the values $F_1(0) = 0.8319081\dots$, $F_2(0) = 0.96937\dots$, $F_4(0) = 0.99857\dots$ of the distribution functions at the origin.

For each family and size we record the sample mean μ_V , standard deviation σ_V , skewness, and excess kurtosis of the largest new eigenvalue λ , the Kolmogorov-Smirnov distance between the standardized sample and each of $\widetilde{TW}_1, \widetilde{TW}_2, \widetilde{TW}_4$ and the standard normal, the fraction of samples below the Ramanujan threshold, and the fraction below the sample mean. The last statistic is MNS's discriminator: the standardized Tracy-Widom laws are nearly indistinguishable by their overall shape, however their masses to the left of the mean, listed in (2.2), differ in the third decimal, and with 10^6 to 5×10^6 samples the binomial standard error is between 2×10^{-4} and 5×10^{-4} , separating the four candidates by tens of standard errors.

The relevance of the ratio $g_V = (\rho - \mu_V)/\sigma_V$ of the mean gap to the standard deviation, with ρ the family's threshold, is the following elementary observation of MNS. For a d -regular family, the probability of being one-sided Ramanujan is

$$P\left(\lambda \leq 2\sqrt{d-1}\right) = P\left(\frac{\lambda - \mu_V}{\sigma_V} \leq g_V\right),$$

and so if the standardized law converges to $\widetilde{TW}_1$, then

$$(2.3) \quad \lim_{V \rightarrow \infty} P\left(\lambda \leq 2\sqrt{d-1}\right) = \begin{cases} F_1(\mu_{TW_1}) = 0.519652\dots & \text{if } g_V \rightarrow 0 \\ \widetilde{F}_1(r) & \text{if } g_V \rightarrow r \in (0, \infty) . \\ 1 & \text{if } g_V \rightarrow \infty \end{cases}$$

MNS’s power law fits pointed to the first case. The Tracy-Widom scaling (1.2), with no shift in the centering, forces the middle case with $r = |\mu_{TW_1}|/\sigma_{TW_1}$, where $\bar{F}_1(r) = F_1(0)$, and this is what our data shows and what Theorem 4 now proves.

2.3. The Dataset and the Pipeline. The dataset consists of 1.4×10^8 samples across ten series, each sample the largest new eigenvalue of one independently generated graph or cover, all with base degree $d = 4$.

family	sizes	samples	model
simple 4-regular	$100 \leq V \leq 20,000$ (8)	5,000,000	two permutations, simple
simple 4-regular (legacy)	$5 \times 10^4 \leq V \leq 5 \times 10^5$ (4)	100,000	two permutations, simple
permutation multigraph	$5 \times 10^4 \leq V \leq 5 \times 10^5$ (4)	100,000	loops and multiple edges kept
$\mathbb{Z}_k$ covers, $k = 3, 4, 5$	$100 \leq n \leq 5000$ (6 each)	1,000,000	random simple 4-regular base
quaternion covers	$100 \leq n \leq 5000$ (6)	1,000,000	random simple 4-regular base
covers of K_5	$100 \leq V \leq 10,000$ (7)	5,000,000	fixed base K_5
covers of $K_5 - e$	$100 \leq V \leq 10,000$ (7)	5,000,000	fixed irregular base
covers of $K_4 - e$	$100 \leq V \leq 10,000$ (7)	1,000,000	fixed base of cycle rank two

The simple 4-regular model draws two independent uniform permutations π_1, π_2 of $\{1, \dots, V\}$ and forms the graph with adjacency matrix $P_1 + P_1^T + P_2 + P_2^T$, resampling locally until the result has no loops or multiple edges, and discarding the rare disconnected samples. This is the permutation model, analyzed first by Friedman [25] (MNS’s family $G_{V,4}$), conditioned on simplicity, equivalently a random degree V cover of the bouquet of two loops. Conditioned on simplicity it is contiguous with the uniform simple 4-regular model, see [36] and the survey [72], and so Theorem 4 applies. The permutation multigraph series keeps the loops and multiple edges, with a loop contributing 2 to the diagonal. The abelian series draw a fresh random simple 4-regular base graph on n vertices for each batch, assign each base edge an independent uniform voltage in $\mathbb{Z}_k$, and record the largest new eigenvalue of the explicit kn -vertex cover, ten covers per base graph. A one way analysis of variance across the ten covers sharing a base gives intraclass correlations below 3×10^{-4} in absolute value at the largest size, and below 2.4×10^{-3} at every size, and so the residual randomness of the base is invisible at the fluctuation scale and the samples may be treated as independent. The quaternion series assigns each edge of a random simple 4-regular base an independent uniform element of Q_8 and records the top eigenvalue of the resulting $4n$ dimensional real block. The fixed base series assign each edge of K_5 , of K_5 minus an edge, and of K_4 minus an edge an independent uniform permutation in S_k , and record the largest new eigenvalue of the cover, with the lifted base eigenvalues removed exactly. These families are analyzed in section 5.

We compute eigenvalues with the implicitly restarted Lanczos method (ARPACK), applied to the adjacency matrix shifted by a small multiple of the identity so that the algebraically largest eigenvalues dominate in magnitude. We remove old eigenvalues by comparing against the base spectrum, computed exactly, with a tolerance of 10^{-10} . Removal by value also discards the duplicated base spectrum of the rare disconnected cover, and so the cover families are conditioned on connectedness, like the simple family (see section 5.2). For the regular families this removes the trivial eigenvalue d and yields λ_2 . The number of computed eigenvalues per graph starts at 2 and grows automatically whenever every computed eigenvalue is old. Two precision caveats apply. The legacy series were generated years earlier in MATLAB and are recorded on a 10^{-12} grid, coarser than machine precision. A per process seeding scheme caused seed collisions between worker tasks in the abelian series: of its 1.8×10^6 base graph batches, 452,508 (25%) are redundant copies of another batch, concentrated at the smallest sizes (55 to 81%

of batches at $n \in \{100, 200\}$, and below 0.1% at $n \geq 1000$), so that the effective number of independent samples at the smallest sizes is smaller than the nominal 10^6 . Duplicated batches agree to better than 3×10^{-13} , and dropping them moves every statistic quoted in this paper by at most two units in its final displayed digit, consistent with the sampling fluctuation of the removed samples. We verified every summary statistic by independent recomputation from the raw samples. The complete dataset, the generation code, and the analysis pipeline that produces every figure and table of this paper are available at <https://github.com/enaslund/random-graphs>.

3. THE REGULAR GRAPH CONTROL EXPERIMENT

In this section we test the pipeline against Theorem 4 along the two simple 4-regular series, diagnose the finite size artifact behind MNS's 52%, and isolate a model, the permutation multigraph, which falls outside the universality class.

3.1. Convergence to the Proven Law. Figure 3.1 shows the standardized empirical density of 5×10^6 samples at $V = 20,000$ against the four standardized candidate laws, and Figure 3.2 the quantile-quantile plot of the largest legacy dataset, $V = 500,000$, against $\widetilde{TW}_1$ across the probability range 10^{-4} to 0.999. The Kolmogorov-Smirnov distances in Figure 3.3 quantify the picture across all sizes: the distance to $\widetilde{TW}_1$ falls with V to the sampling floor, reaching 8.7×10^{-4} at $V = 20,000$ (the floor at 5×10^6 samples is roughly 4×10^{-4}), while the distances to $\widetilde{TW}_2$, $\widetilde{TW}_4$, and the normal stall at 3.9×10^{-3} , 7.8×10^{-3} , and 1.9×10^{-2} respectively. At the legacy sizes the sample is statistically indistinguishable from the law: at $V = 500,000$ the Kolmogorov-Smirnov test accepts $\widetilde{TW}_1$ with $p = 0.99$. The mass to the left of the mean, MNS's decisive statistic, reaches 0.51885 at $V = 20,000$ and 0.52011 at $V = 500,000$ against the TW_1 value 0.519652 from (2.2), with the competing values 0.515016, 0.511069, and 0.5 excluded by more than fifteen binomial standard errors (see Figure 4.3 in section 4). The sample skewness and excess kurtosis at the largest sizes, 0.286 and 0.156 at $V = 20,000$, converge to the TW_1 values 0.293 and 0.165 (Figure 4.4, also in section 4).

The scaling constants of (1.2) are confirmed as well. For $d = 4$ we have that $c(4) = 3^{-1/6} = 0.8326831\dots$, and so the conjectured, now proven, amplitudes are

$$(3.1) \quad 2\sqrt{3} - \mu_V \approx 1.0046602 V^{-2/3}, \quad \sigma_V \approx 1.0558282 V^{-2/3}.$$

Figure 3.4 shows the mean gap and standard deviation against V on logarithmic axes with the $V^{-2/3}$ guide, and Figure 3.5 shows the compensated quantities $(2\sqrt{3} - \mu_V) V^{2/3}$ and $\sigma_V V^{2/3}$, which approach the two constants of (3.1): at $V = 500,000$ the measured standard deviation is 1.685×10^{-4} against the predicted 1.676×10^{-4} , and the measured gap is 1.684×10^{-4} against the predicted 1.595×10^{-4} , the mean converging visibly more slowly than the standard deviation. The constants (3.1) appear in Sodin's work [67] on a closely related model in which the matrix entries carry independent random signs. We note that the arXiv posting of [67] was withdrawn over an error in one of its lemmas, and the published version carries a 2017 erratum, so that Theorem 4 is the rigorous source for (1.2).

3.2. The Approach to the Limit. The theorems give no rate of convergence, and the data is precise enough to measure one. Rates at the Tracy-Widom edge have a history in random matrix theory: Choup [22] computed the Edgeworth expansion of the largest eigenvalue law of the finite GOE, Johnstone and Ma [48] proved that the rate of approach depends on the centering convention, and rates of order $N^{-1/3}$ are proven

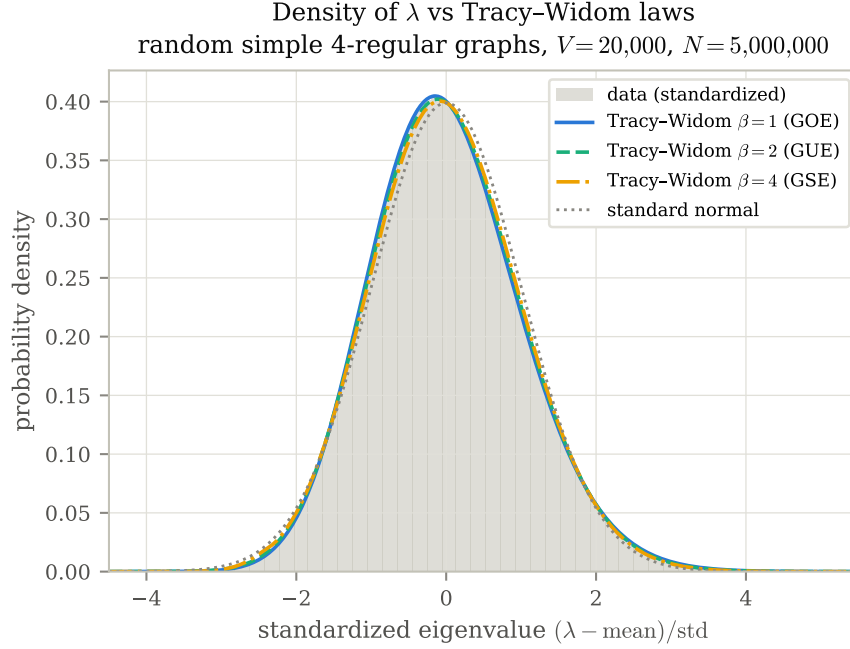


FIGURE 3.1. Standardized empirical density of the largest non-trivial eigenvalue over 5×10^6 random simple 4-regular graphs on $V = 20,000$ vertices, against the standardized Tracy-Widom $\beta = 1, 2, 4$ densities and the standard normal.

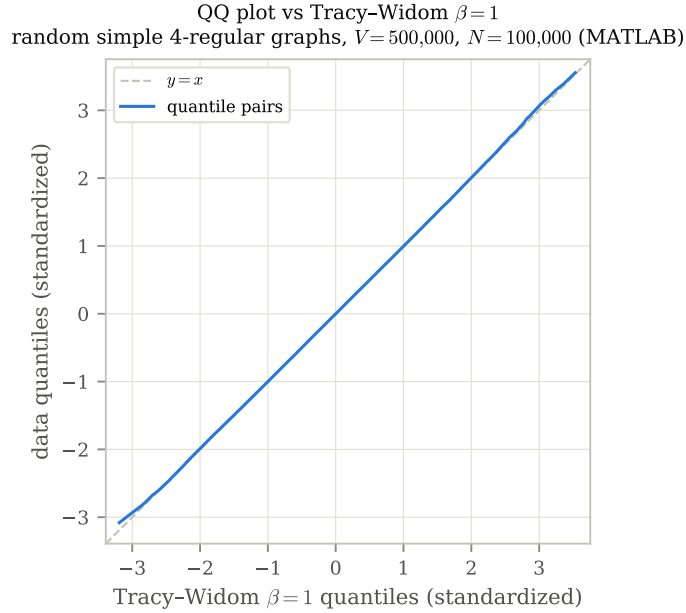


FIGURE 3.2. Quantile-quantile plot of 10^5 samples at $V = 500,000$ against $\widehat{TW}_1$, probability range 10^{-4} to 0.999.

for Wigner matrices [66] and sparse Erdős-Rényi graphs [19]. For random regular graphs no rate is proven. Write F_V for the distribution function of the standardized largest non-trivial eigenvalue and $\widetilde{F}_1, \widetilde{f}_1$ for the standardized Tracy-Widom distribution and

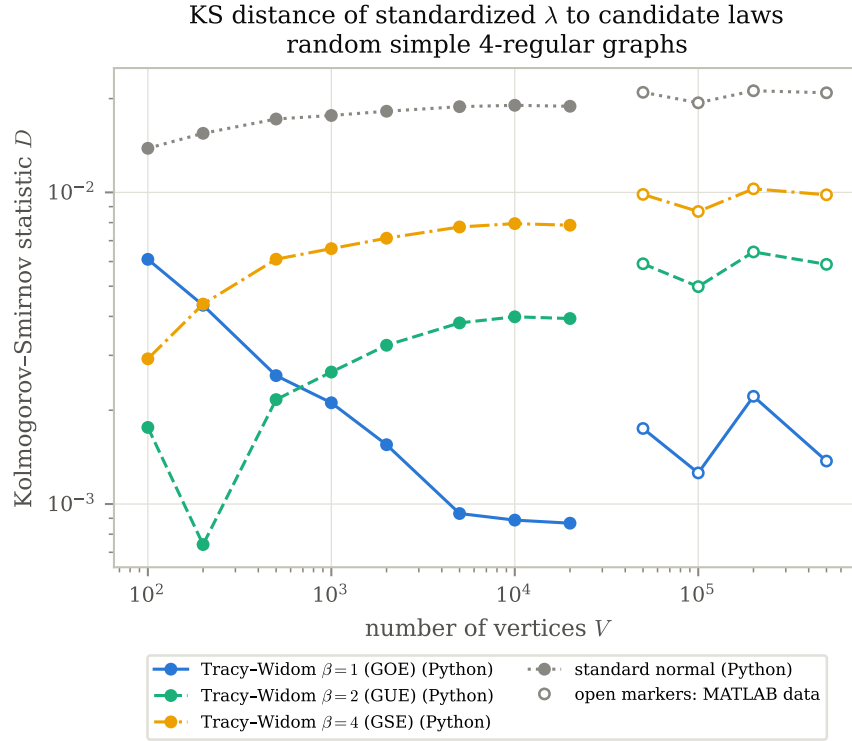


FIGURE 3.3. Kolmogorov-Smirnov distance of the standardized samples to the standardized Tracy-Widom $\beta = 1, 2, 4$ laws and the standard normal, for random simple 4-regular graphs (filled markers: 5×10^6 samples per size, open markers: legacy data with 10^5 samples). Only the distance to $\beta = 1$ keeps falling.

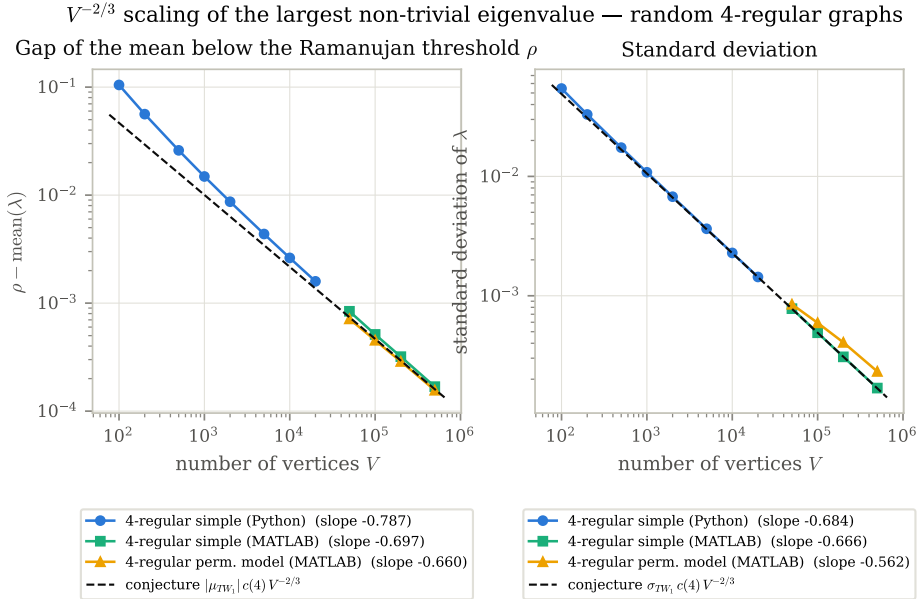


FIGURE 3.4. Mean gap $2\sqrt{3} - \mu_V$ and standard deviation against V for the three 4-regular series, with fitted slopes and the $V^{-2/3}$ Tracy-Widom line.

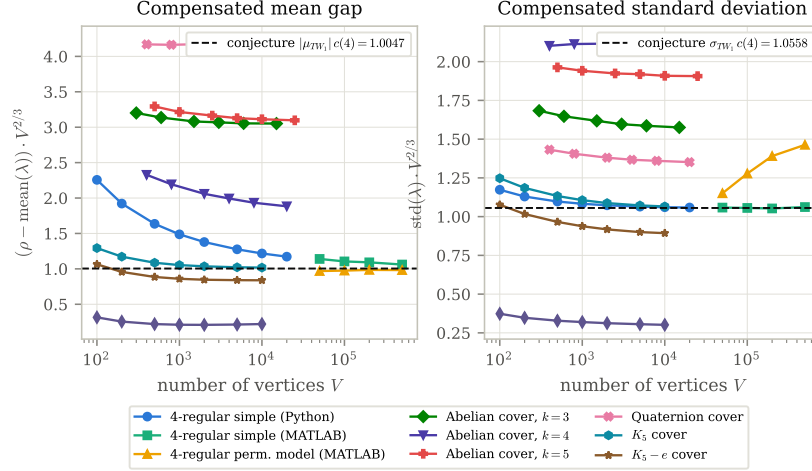
Compensated $V^{2/3}$ amplitudes — dashed: constants conjectured for random 4-regular graphs

FIGURE 3.5. Compensated scaling: $(\rho - \mu_V) V^{2/3}$ and $\sigma_V V^{2/3}$ against V for all families, where ρ is the family's Ramanujan threshold. The dashed lines are the regular graph amplitudes (3.1), which the simple 4-regular and K_5 cover series approach, while each remaining cover family plateaus at its own amplitude, so the $V^{-2/3}$ rate is universal while the constant is family dependent (see sections 4 and 5).

density. Across the simple 4-regular family the deviation $F_V - \tilde{F}_1$ is, to within 6 to 14 percent of its squared norm, proportional to the single fixed profile $(t^2 - 1) \tilde{f}_1(t)$, a correction of Edgeworth type signaling a skewness deficit, and so to the resolution of 5×10^6 samples the finite size law is the one parameter family

$$F_V(t) \approx \tilde{F}_1(t) - \hat{c}(V) (t^2 - 1) \tilde{f}_1(t),$$

where the fitted amplitude $\hat{c}(V)$ decays like a power law with exponent -0.46 ± 0.04 , and the K_5 cover family of section 5 lies on the same curve (Figure 3.6). The same parameter accounts for the residuals of the individual statistics: the mass left of the mean at $V = 20,000$ still sits -8.0×10^{-4} from the TW_1 value, 3.6 binomial standard errors, against an Edgeworth prediction of $-\hat{c} \tilde{f}_1(0) = -5.3 \times 10^{-4}$, and after subtracting the fitted profile the left tails of the largest datasets agree with $\tilde{TW}_1$ to within three standard errors down to probability 6×10^{-7} . The corrections to the amplitudes decay faster: fitting $A + BV^{-\theta}$ to the compensated standard deviation of the simple family gives $A = 1.0548$ with $\theta = 0.63$ freely fitted and $A = 1.0562$ with θ fixed at $2/3$, bracketing the conjectured $\sigma_{TW_1}c(4) = 1.05583$ of (3.1) to within a tenth of a percent, while the compensated mean gap admits no single correction exponent over our range, which is why it converges visibly more slowly in Figure 3.5.

3.3. The Origin of the 52% Conjecture. MNS reached the first case of (2.3) by fitting single power laws to the mean gap and standard deviation over their full range of sizes. Our data reproduces their fit and shows exactly how it misleads. Fitting one line to $\log(2\sqrt{3} - \mu_V)$ against $\log V$ over our eight sizes $V \leq 20,000$, the MNS range, gives a slope of -0.787 , while the same fit to $\log \sigma_V$ gives -0.684 : a mean exponent apparently steeper than the standard deviation exponent, exactly the pattern from which MNS concluded $g_V \rightarrow 0$ and hence 52%, with numbers closely matching their

The deviation of the standardized ECDF from TW_1 is a single skewness profile with power-law amplitude

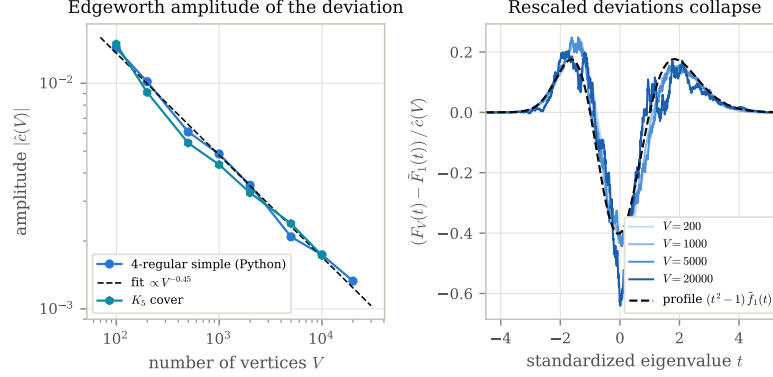


FIGURE 3.6. The approach to the limit. Left: the amplitude $\hat{c}(V)$ of the deviation of the standardized empirical distribution from $\widetilde{TW}_1$, for the simple 4-regular and K_5 cover families, with the fitted power law. Right: the deviations at four sizes, rescaled by $\hat{c}(V)$, collapse onto the fixed profile $(t^2 - 1)\tilde{f}_1(t)$ of a skewness deficit.

$d = 3$ values $m \approx -0.79$ and $s \approx -0.72$. The local exponents tell a different story. Figure 3.7 plots the slope between consecutive sizes: the standard deviation exponent sits within 0.01 of $-2/3$ from $V \approx 2000$ onward, while the mean gap exponent drifts monotonically from -0.90 at $V \approx 100$ to -0.72 at $V = 20,000$ within the five million sample series, with the legacy series continuing the descent and reaching -0.70 over its last pair at $V = 500,000$, still short of $-2/3$. A single fitted line over small sizes averages this drift and reports a transient. MNS saw the warning sign, observing that their best fit exponents increased monotonically as sizes were added, and were appropriately cautious. At $V \leq 20,000$ the data could do no better. The ratio g_V resolves the question directly: as Figure 4.7 shows, g_V does not tend to 0 for any family, and for the simple 4-regular series it descends to 0.9993 at $V = 500,000$, approaching the Tracy-Widom ratio $|\mu_{TW_1}|/\sigma_{TW_1} = 0.951537\dots$ from above, in agreement with the middle case of (2.3) and with $F_1(0) = 0.8319081\dots$ as the limiting Ramanujan probability.

3.4. Failure of the Multigraph Model. MNS pooled models with and without loops and multiple edges, observing similar behavior at their sample sizes. At ours, the difference is qualitative. The permutation multigraph series, identical to the legacy simple series except that loops and multiple edges are kept, has Kolmogorov-Smirnov distance to every candidate law growing with V , reaching 0.085 at $V = 500,000$, sample skewness 74.5, and excess kurtosis 1.15×10^4 , against the TW_1 values 0.293 and 0.165. Figure 3.8 shows the mechanism: the right tail of the simple model dies within 10^{-3} of the threshold $2\sqrt{3}$, while the multigraph model has a heavy tail of far outliers, reaching $\lambda = 3.538$ at $V = 50,000$, with roughly 2×10^{-5} of its mass more than ten standard deviations above the threshold at $V = 500,000$ and an extreme sample 155 standard deviations above the mean. These outliers are localized eigenvalues seeded by loops and multiple edges, whose density vanishes in the limit but whose spectral effect does not vanish at the $V^{-2/3}$ fluctuation scale. In the language of section 5.2, loops and multiple edges are binding defects of cycle rank one, and so they occur at constant rate per graph. The lesson for experiments in this area is that the universality class is a property of the conditioned model: the simple and multigraph models share the same

Local log-log slopes between consecutive sizes — a single power-law fit over small V overestimates the gap exponent

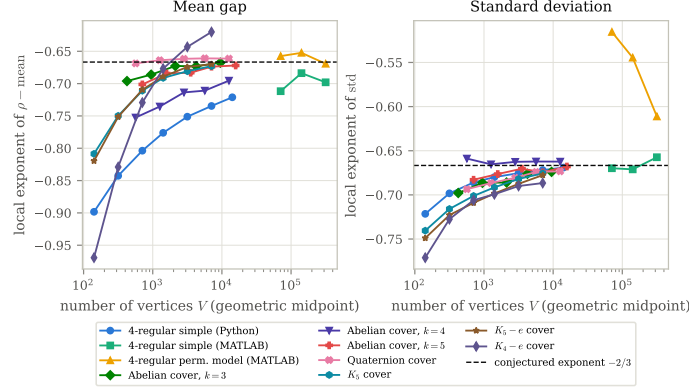


FIGURE 3.7. Local (consecutive size) log-log exponents of the mean gap and the standard deviation for all series, with the $-2/3$ line. The mean gap exponent approaches $-2/3$ from below as V grows, and so a single power law fitted over small sizes overestimates its steepness, wrongly suggesting that the gap outruns the standard deviation. This is the origin of the 52% conjecture.

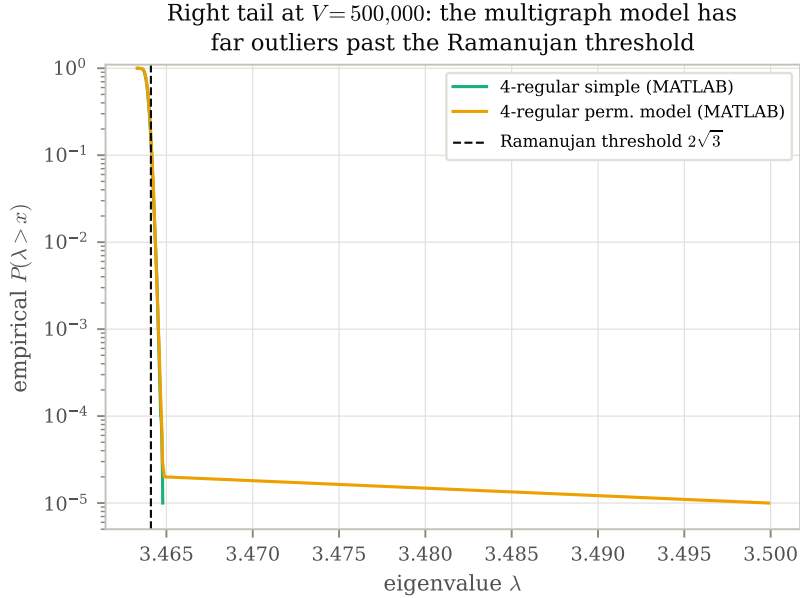


FIGURE 3.8. Right tail $P(\lambda > x)$ at $V = 500,000$ for the simple and multigraph permutation models on logarithmic scale. The simple model's tail terminates within 10^{-3} of $2\sqrt{3}$, while the multigraph model's outliers, localized eigenvalues from loops and multiple edges, reach far beyond the threshold.

limiting spectral distribution and the same almost sure convergence of λ_2 to $2\sqrt{3}$, and are separated only at the fluctuation scale, where one is Tracy-Widom and the other is not even tight around the threshold in units of its own standard deviation, its ratio g_V falling through 0.67 with no sign of stabilizing.

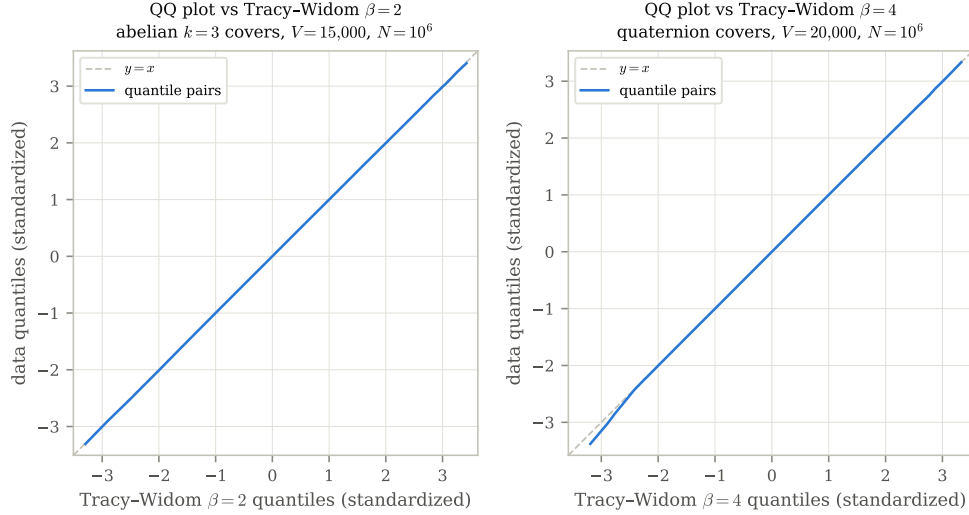


FIGURE 4.1. Quantile-quantile plots of the largest $\mathbb{Z}_3$ cover dataset against Tracy-Widom $\beta = 2$ (left) and of the largest quaternion cover dataset against $\beta = 4$ (right), both standardized, probability range 10^{-4} to 0.999.

4. COVERS AND THE THREEFOLD WAY

We now turn to the main experiment. Each abelian and quaternion family isolates a definite set of isotypic blocks, and Conjecture 1 predicts the symmetry class of each: $\mathbb{Z}_3$ covers should follow $\widetilde{TW}_2$, quaternion covers $\widetilde{TW}_4$, the regular graph families of section 3 $\widetilde{TW}_1$, and the mixed families $\mathbb{Z}_4$ and $\mathbb{Z}_5$ no single Tracy-Widom law at all.

4.1. Ensemble Selection in the Pure Families. Figure 4.2 is the structural result of this paper: for each pure family, the Kolmogorov-Smirnov distance to the law of its own symmetry class falls to the sampling floor while the distance to $\widetilde{TW}_1$ (dotted) stalls. At the largest sizes, the $\mathbb{Z}_3$ family reaches distance 1.0×10^{-3} from $\widetilde{TW}_2$ (floor at 10^6 samples: roughly 9×10^{-4}) while its distance to $\widetilde{TW}_1$ is stuck at 5.4×10^{-3} , and the quaternion family reaches 7.5×10^{-4} from $\widetilde{TW}_4$ with distance 8.8×10^{-3} to $\widetilde{TW}_1$. The mass to the left of the mean (Figure 4.3) separates the classes at higher resolution: the $\mathbb{Z}_3$ series converges to 0.5145 against the TW_2 value 0.515016, the quaternion series to 0.5109 against the TW_4 value 0.511069, and the regular series to 0.5201 against the TW_1 value 0.519652, each matching its own class, with every competing class excluded by six or more binomial standard errors for the cover families and by three or more for the legacy regular sample. The affine invariant moments agree: Figure 4.4 shows the sample skewness and excess kurtosis of each family converging to the values of its own class, (0.213, 0.093) at the largest $\mathbb{Z}_3$ size against the TW_2 values (0.224, 0.093), and (0.156, 0.042) at the largest quaternion size against the TW_4 values (0.166, 0.049). Figure 4.1 gives the quantile level view of the same selection, plotting the $\mathbb{Z}_3$ sample against the $\beta = 2$ quantiles and the quaternion sample against the $\beta = 4$ quantiles.

Figure 4.5 overlays the standardized densities of the largest datasets from five families on the $\widetilde{TW}_1$ density: at the resolution of the eye all collapse to the Tracy-Widom shape, and the discrimination between the β classes lives in the statistics above.

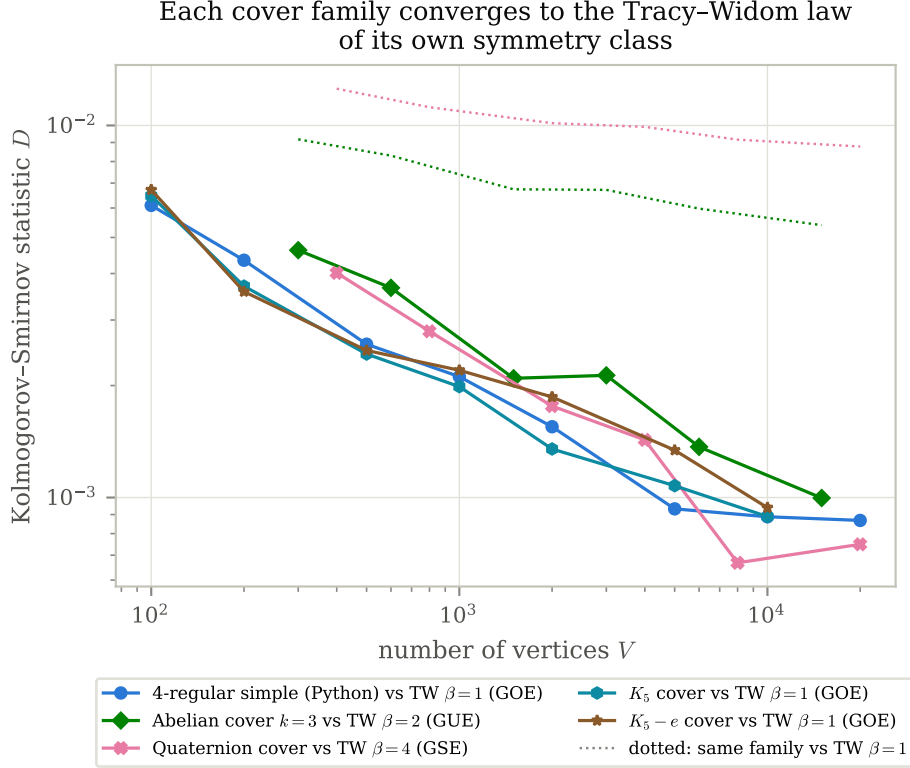


FIGURE 4.2. Kolmogorov-Smirnov distance of each pure family to the Tracy-Widom law of its own symmetry class (solid): regular graphs and the fixed base K_5 and $K_5 - e$ cover families to $\beta = 1$, $\mathbb{Z}_3$ covers to $\beta = 2$, quaternion covers to $\beta = 4$, with the distance to $\beta = 1$ dotted for contrast. The matching class distances fall to the sampling noise floor. The mixed families $\mathbb{Z}_4$ and $\mathbb{Z}_5$ match no single law.

4.2. The Mixed Families and the Maximum Law. The largest new eigenvalue of a $\mathbb{Z}_4$ cover is the maximum of a GOE class maximum and a GUE class maximum, and that of a $\mathbb{Z}_5$ cover is the maximum of two GUE class maxima, and so under Conjectures 1 and 2 neither should converge to any single Tracy-Widom law. This is visible in the raw statistics: the Kolmogorov-Smirnov distance of the $\mathbb{Z}_4$ family to every candidate law grows with V , reaching 1.9×10^{-2} against $\widehat{TW}_1$ at the largest size, and the $\mathbb{Z}_5$ family, nearest $\widehat{TW}_1$ at moderate sizes, drifts away from it as V grows.

The maximum structure can be tested quantitatively. If the two blocks of a family fluctuate independently with the same centering $2\sqrt{d-1}$ and the same scale constant, then the standardized largest new eigenvalue converges to the standardized law of

$$M = \max(X_1, X_2),$$

where X_1, X_2 are independent Tracy-Widom variables of the appropriate classes on a common scale. Computing the distribution function $F_1 F_2$ (for $\mathbb{Z}_4$) and F_2^2 (for $\mathbb{Z}_5$) numerically gives the predicted constants in Table 1: the ratio $|\mu_M|/\sigma_M$, the mass of M left of its mean, and the Ramanujan functional $F_{\beta_1}(0)F_{\beta_2}(0)$. The $\mathbb{Z}_5$ family, whose two blocks have identical marginal distributions and for which the equal scale hypothesis is automatic, matches the mass and Ramanujan predictions to three decimal places at $n = 5000$, and the ratio to within 3×10^{-3} . The $\mathbb{Z}_4$ family, whose blocks belong to

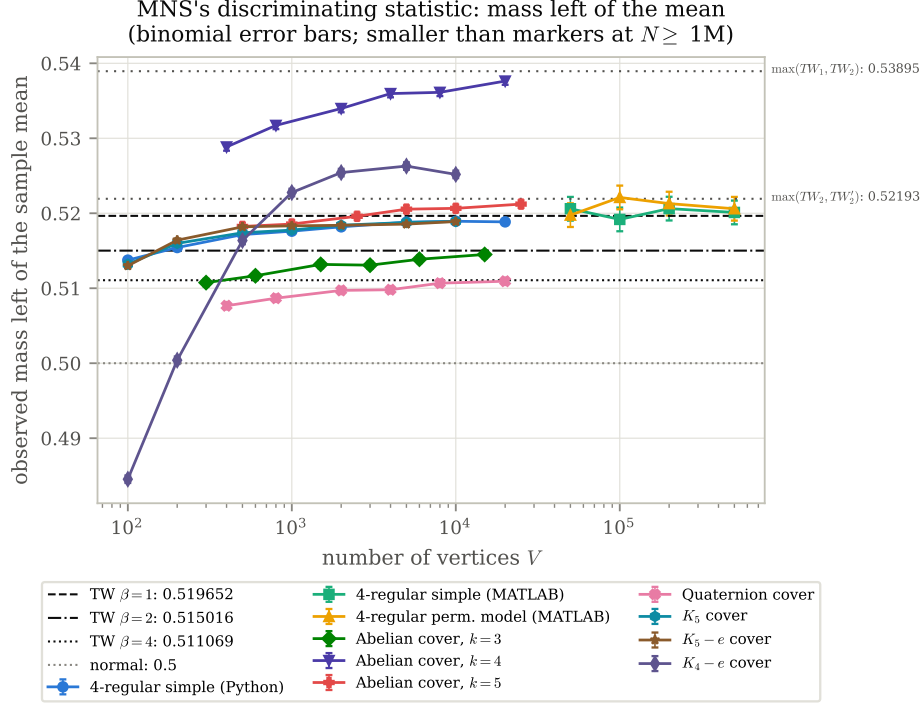


FIGURE 4.3. The fraction of samples below the sample mean against V for all families, with the Tracy-Widom $\beta = 1, 2, 4$ and normal reference values. This replicates MNS's decisive experiment at fifty times their sample size: with 10^6 to 5×10^6 samples the binomial standard error is between 2×10^{-4} and 5×10^{-4} and the four candidate laws are separated by tens of standard errors. Each pure family selects the value of its own symmetry class, while the mixed family $\mathbb{Z}_4$ drifts to ≈ 0.538 , consistent with the maximum law of Table 1 and no Tracy-Widom law.

Third and fourth moments converge to the Tracy-Widom values of each family's symmetry class (permutation model omitted: kurtosis up to $\sim 10^4$)

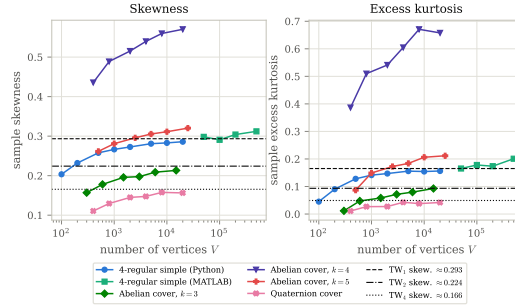


FIGURE 4.4. Sample skewness and excess kurtosis against V with the Tracy-Widom $\beta = 1, 2, 4$ reference values. The regular families converge to the $\beta = 1$ moments, $\mathbb{Z}_3$ to $\beta = 2$, and quaternion to $\beta = 4$. The multigraph model (excess kurtosis up to 10^4) and the fixed base cover families, whose third and fourth moments are dominated by the far outlier tails of section 5, are omitted.

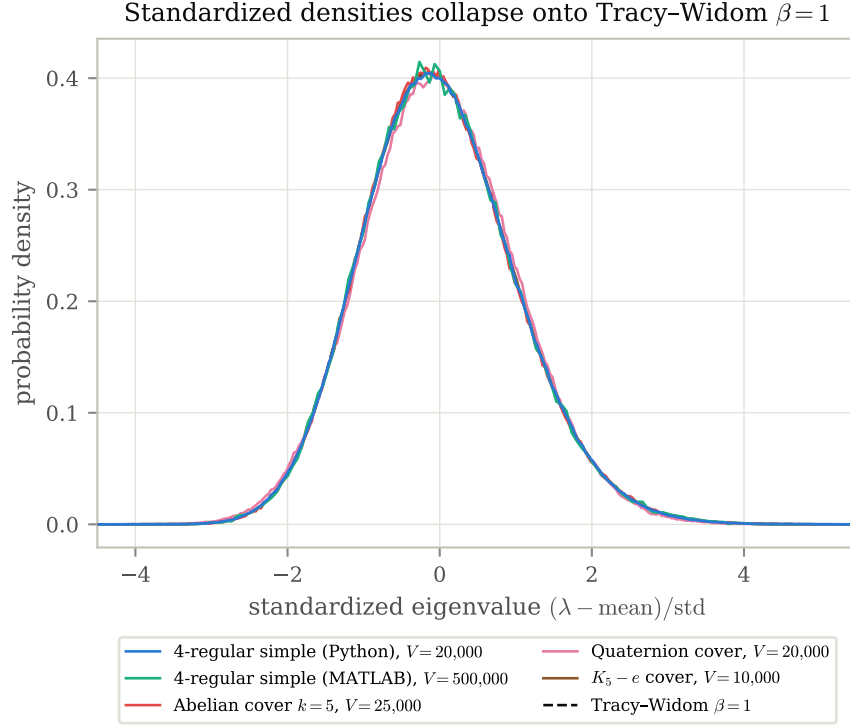


FIGURE 4.5. Standardized empirical densities of the largest datasets from five families overlaid on the standardized $\beta = 1$ Tracy-Widom density. The β classes are indistinguishable at this resolution, and are separated by the statistics of Figures 4.2, 4.3, and 4.4.

different classes and converge at visibly different rates, is still drifting at $n = 5000$, in each statistic monotonically toward the predicted value. The maximum structure is visible at the moment level as well: at $n = 5000$ the $\mathbb{Z}_4$ skewness and excess kurtosis are $(0.570, 0.658)$ against the $\max(TW_1, TW_2)$ values $(0.584, 0.670)$, and the $\mathbb{Z}_5$ values are $(0.320, 0.212)$ against $(0.329, 0.225)$, in each case drifting monotonically toward the prediction. Note that the drift away from the pure laws is invisible at moderate sizes: with 10^6 samples at $V = 2500$ the Kolmogorov-Smirnov test accepts $\widetilde{TW}_1$ for the $\mathbb{Z}_5$ family ($p = 0.81$), and only the larger sizes reveal that the distance is growing, a caution for experiments at the scale of MNS.

The Ramanujan column of Table 1 is the experimental form of Conjecture 2. Each family approaches the product $\prod F_\beta(0)$ over its blocks: one GUE block gives $F_2(0) = 0.96937\dots$ for $\mathbb{Z}_3$, two give $F_2(0)^2 = 0.93968\dots$ for $\mathbb{Z}_5$, matched by the observed 0.94011 at $n = 5000$, the mixed $\mathbb{Z}_4$ product $F_1(0)F_2(0) = 0.80643\dots$ is approached from above through 0.82293, and the quaternion block gives $F_4(0) = 0.99857\dots$ against the observed 0.99825. Figure 4.6 shows the full curves, and Figure 4.7 the underlying ratios g_V , each family flattening onto the ratio of its own limit law. In particular the limiting probability of being Ramanujan is not one number but a representation theoretic invariant of the family, always strictly between 0 and 1.

4.3. Scaling Amplitudes. All ten series share the $V^{-2/3}$ rate, with family dependent amplitudes (Figure 3.5). The fixed base families are treated in section 5. For the abelian families the amplitudes themselves appear to carry no new constant: each character block A_χ is the adjacency matrix of the random base graph with its entries rotated by

family	limit law	g_V	mass left of mean	$P(\text{Ramanujan})$
simple 4-regular	TW_1	0.9993 (0.9515)	0.52011 (0.51965)	0.84352 (0.83191)
$\mathbb{Z}_3$ covers	TW_2	1.9369 (1.9640)	0.51450 (0.51502)	0.96793 (0.96937)
$\mathbb{Z}_4$ covers	$\max(TW_1, TW_2)$	0.8811 (0.8099)	0.53763 (0.53895)	0.82293 (0.80643)
$\mathbb{Z}_5$ covers	$\max(TW_2, TW'_2)$	1.6245 (1.6219)	0.52122 (0.52193)	0.94011 (0.93968)
quaternion covers	TW_4	3.1263 (3.2061)	0.51094 (0.51107)	0.99825 (0.99857)
K_5 covers	TW_1	0.9551 (0.9515)	0.51891 (0.51965)	0.83281 (0.83191)
$K_5 - e$ covers	TW_1	0.9387 (0.9515)	0.51890 (0.51965)	0.82916 (0.83191)
$K_4 - e$ covers	TW_1	0.7326 (0.9515)	0.52518 (0.51965)	0.79490 (0.83191)

TABLE 1. Observed statistics at the largest available size of each family against the values predicted by Conjectures 1, 2, and 3: the ratio $g_V = (\rho - \mu_V) / \sigma_V$ of the mean gap to the standard deviation, the mass left of the sample mean, and the fraction of samples below the Ramanujan threshold ρ ($\rho = 2\sqrt{3}$ except for the $K_5 - e$ and $K_4 - e$ rows, where ρ is the spectral radius of the universal cover of the base). Predicted values for the maximum laws were computed numerically from the Tracy-Widom distributions assuming independent blocks on a common scale. Each cell shows the observed value with the predicted limit in parentheses. Observed values are at $V = 500,000$ (simple), $n = 5000$ (abelian and quaternion covers), and $V = 10,000$ (the fixed base families). The $K_4 - e$ family converges slowest and its observed values are still far from the limits, for the reasons given in section 5.

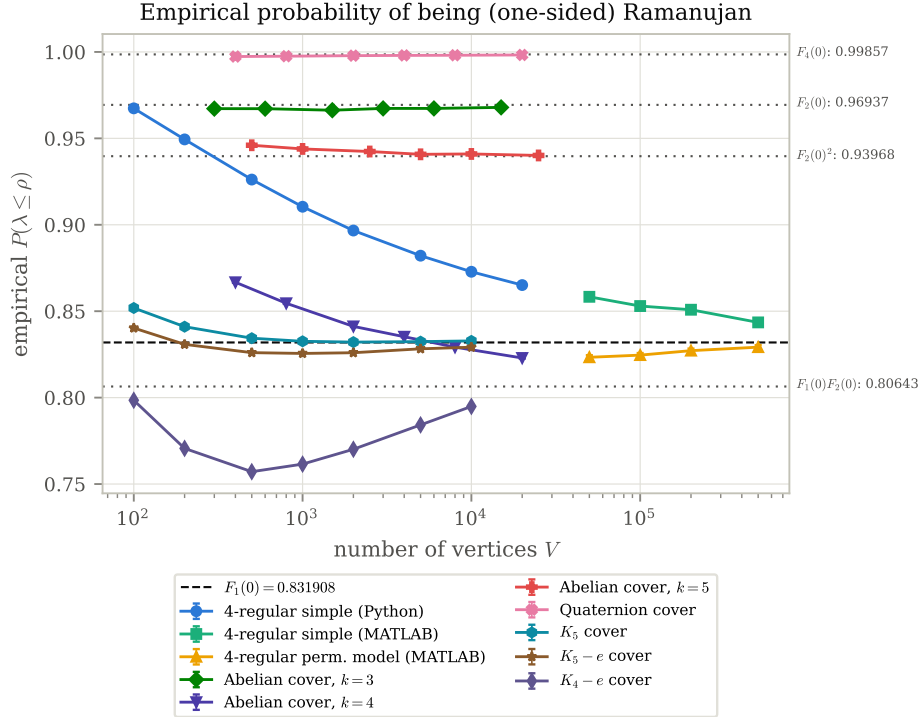


FIGURE 4.6. Empirical $P(\lambda \leq \rho)$ against V for all families, where ρ is the family's Ramanujan threshold, with the limit $F_1(0) = 0.8319081\dots$ for the $\beta = 1$ families. Each cover family approaches the product of Tracy-Widom functionals of Conjecture 2 listed in Table 1.

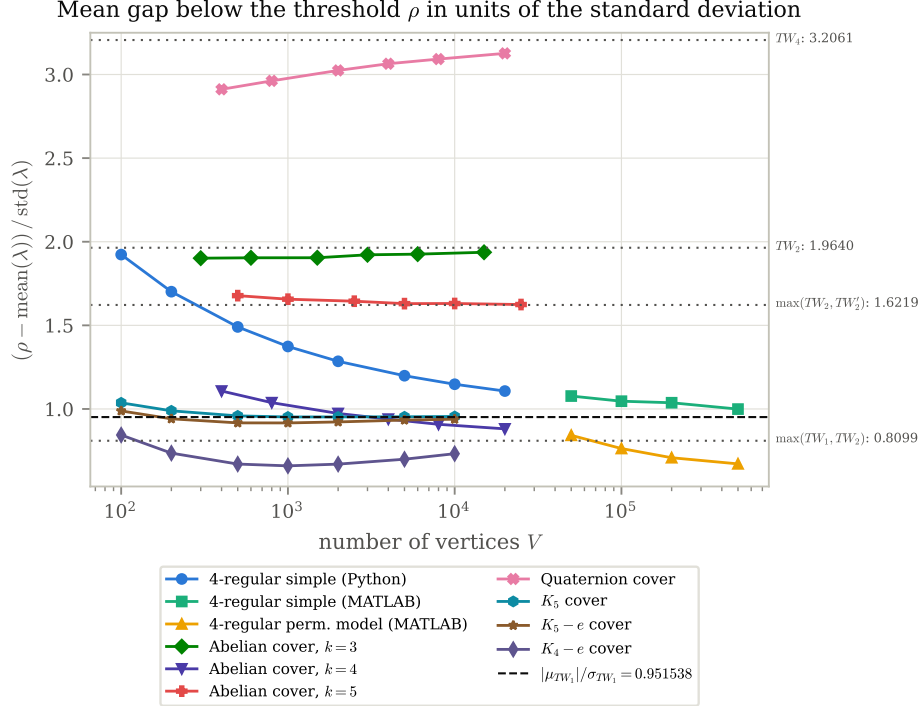


FIGURE 4.7. The ratio $g_V = (\rho - \mu_V) / \sigma_V$ against V for all families, where ρ is the family's Ramanujan threshold, with the $\beta = 1$ reference $|\mu_{TW_1}| / \sigma_{TW_1} = 0.951537\dots$ dashed. MNS conjectured this ratio tends to 0, however each family plateaus at the ratio of its own limit law: 1.96 for $\mathbb{Z}_3$, 1.62 for $\mathbb{Z}_5$, 3.21 for quaternion covers, and the $\beta = 1$ reference for the regular and fixed base families.

the unimodular phases $\chi(\sigma_e)$, a perturbation which does not change the tree-like local structure, and so we conjecture that every non-trivial character block obeys Conjecture 1 with the same constant $c_\chi = c(d)$ as the unphased regular graph, where $c(d)$ is defined in (1.2) and n is the base size. The data supports this to about one percent at $n = 5000$. For $\mathbb{Z}_3$ the prediction $\sigma \approx \sigma_{TW_2} c(4) n^{-2/3} = 2.568 \times 10^{-3}$ compares to the measured 2.590×10^{-3} , and the predicted mean gap $|\mu_{TW_2}| c(4) n^{-2/3} = 5.044 \times 10^{-3}$ to the measured 5.017×10^{-3} . For $\mathbb{Z}_5$, using the maximum law of Table 1, the predicted standard deviation and gap are 2.218×10^{-3} and 3.598×10^{-3} against the measured 2.230×10^{-3} and 3.623×10^{-3} . The quaternion family plateaus at an amplitude a clean factor below the analogous prediction: the measured compensated standard deviation $\sigma_V n^{2/3}$ drifts toward $0.534 \approx \sigma_{TW_4} c(4) 2^{-1/6}$ rather than $\sigma_{TW_4} c(4) = 0.599$, suggesting that the correct effective size for the quaternionic block differs from n by a fixed power of 2, a bookkeeping familiar from the conventions of the GSE. We do not regard the exponent $-1/6$ as established.

5. COVERS OF A FIXED BASE GRAPH

We now fix the base graph and let only the cover vary, testing three bases: the complete graph K_5 , whose covers are 4-regular and calibrate the fixed base model against section 3, the irregular graph $K_5 - e$, and the cycle rank two graph $K_4 - e$ of subsection 5.3. Nothing in Conjecture 3 requires the base graph to be regular, and in the irregular case nothing is proven: the largest new eigenvalue should concentrate at the

spectral radius $\rho(B)$ of the universal cover, an algebraic number lying between $2\sqrt{\delta - 1}$ and $2\sqrt{\Delta - 1}$, where δ and Δ are the minimum and maximum degrees of B , with TW_1 fluctuations at scale $k^{-2/3}$ and limiting Ramanujan probability $F_1(0)$, universally in the base graph. The threshold $\rho(B)$ is also the benchmark of the X -Ramanujan property of Mohanty and O'Donnell [60], who construct infinite families of finite graphs whose second eigenvalue is at most the spectral radius of a fixed infinite cover. It is the right benchmark: by the generalized Alon-Boppana theorems of Friedman and Tillich [33], no infinite family of graphs sharing the universal cover of B has second eigenvalue asymptotically below $\rho(B)$, and Hoory [41] bounds $\rho(B)$ itself below by $2\sqrt{\bar{d} - 1}$, where $\bar{d}$ is the average degree of B .

The main base is $K_5 - e$, the complete graph on five vertices minus an edge, one of the smallest irregular base graphs of minimum degree three, with eigenvalues $1 + \sqrt{7} = 3.6457\dots$, 0 , -1 , -1 , and $1 - \sqrt{7}$. Since the largest old eigenvalue exceeds $\rho(K_5 - e)$, the largest new eigenvalue of a cover is not even its second largest eigenvalue, so that the exact removal of the lifted base spectrum matters. By a result of McKay [58], $\rho(K_5 - e)$ is the largest zero of

$$x^{14} - 20x^{12} + 122x^{10} - 152x^8 - 1295x^6 + 4540x^4 - 5948x^2 - 2000,$$

an irreducible polynomial whose roots admit no expression in radicals, giving

$$\rho(K_5 - e) = 3.26287646593635862827\dots,$$

which we re-derived and verified independently. Note the contrast with the regular case, where $\rho = 2\sqrt{d - 1}$: for a general base even the centering of the conjectured law is a non-trivial algebraic quantity (see [11, 34, 58] on computing ρ , and [23] for related spectral experiments on structured random graphs).

5.1. The Bulk Law. We sampled the K_5 and $K_5 - e$ families at the scale of the strongest regular graph data: 5×10^6 independent uniform covers per size, at cover degrees $20 \leq k \leq 2000$, giving graphs on $V = 5k$ between 100 and 10,000 vertices. The covers of K_5 are 4-regular graphs with Ramanujan threshold $2\sqrt{3}$, of a different fine structure than the permutation model, since the vertex set carries a fixed partition into five independent fibers with edges only between fibers adjacent in K_5 , and so they test whether the limit law and its constants depend on anything beyond the degree. The covers of $K_5 - e$ test Conjecture 3 where no theorem reaches, around the algebraic centering $\rho(K_5 - e)$.

Figure 5.1 shows the outcome of the distributional test. For both families the Kolmogorov-Smirnov distance to $\widetilde{TW}_1$ falls monotonically with V , reaching 8.9×10^{-4} for K_5 and 9.4×10^{-4} for $K_5 - e$ at $V = 10,000$, a small multiple of the sampling floor and equal to the simple 4-regular family's distance at the same size, while the distances to the other laws stall or grow, ending at 4.0×10^{-3} ($\widetilde{TW}_2$), 7.9×10^{-3} ($\widetilde{TW}_4$), and 1.9×10^{-2} (normal) for $K_5 - e$. A two sample comparison is sharper still: the Kolmogorov-Smirnov distance between the standardized simple and K_5 samples at $V = 10,000$ is 3.7×10^{-4} ($p = 0.88$), while each lies 8.9×10^{-4} from $\widetilde{TW}_1$ ($p \approx 10^{-3}$), and so at 10^7 combined samples the two models are statistically indistinguishable from one another while both still differ measurably from the limit. At small sizes the families are well separated ($p = 2 \times 10^{-15}$ at $V = 100$), merging from $V \approx 5000$ on: the limit is universal, and the finite size path to it becomes universal as well. Figure 5.2 shows the standardized density of the largest $K_5 - e$ dataset, and Figure 5.3 the corresponding quantile plot. The mass left of the mean reaches 0.51891 (K_5) and 0.51890

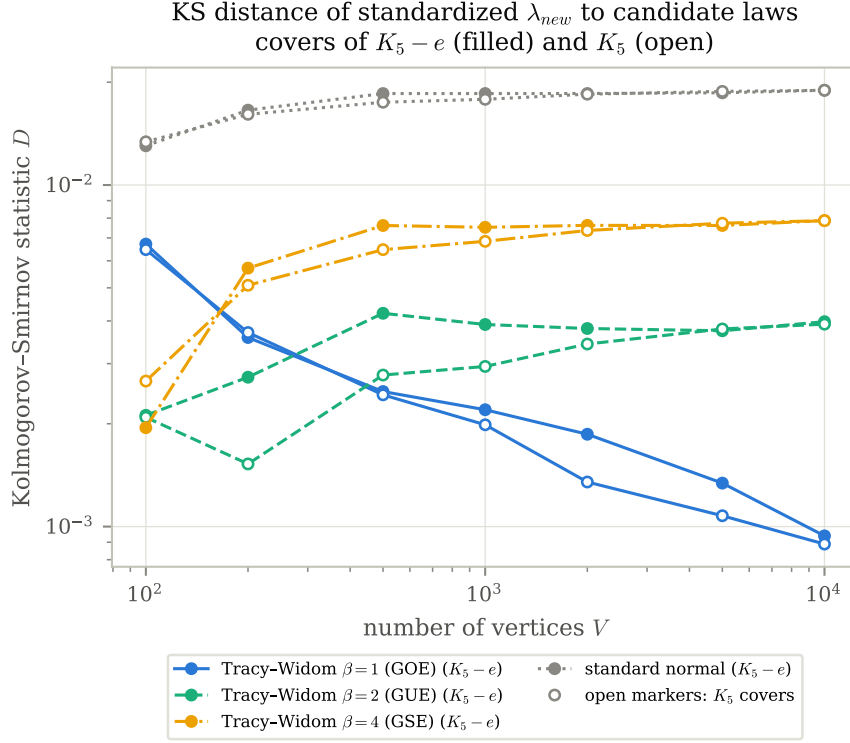


FIGURE 5.1. Kolmogorov-Smirnov distance of the standardized largest new eigenvalue to the four candidate laws for covers of $K_5 - e$ (filled markers, standardized around $\rho(K_5 - e)$) and of K_5 (open markers), 5×10^6 samples per size. For both families only the distance to $\beta = 1$ keeps falling.

($K_5 - e$) against the TW_1 value 0.519652, on the same convergence path as the simple family at equal sizes, and the ratio g_V reaches 0.9551 and 0.9387 against the TW_1 ratio $|\mu_{TW_1}|/\sigma_{TW_1}$ (Figure 4.7). The local exponents of the mean gap and standard deviation descend to -0.673 and -0.676 for K_5 and -0.670 and -0.678 for $K_5 - e$ over the last pair of sizes, approaching $-2/3$ exactly as for the regular families (Figure 3.7). The empirical Ramanujan probabilities at $V = 10,000$ are

$$P(\lambda_{new} \leq 2\sqrt{3}) = 0.83281, \quad P(\lambda_{new} \leq \rho(K_5 - e)) = 0.82916,$$

against the conjectured $F_1(0) = 0.8319081 \dots$ (Table 1). Every statistic selects the $\beta = 1$ class, as the reality of the standard representation predicts, and for $K_5 - e$ it does so around a centering which is not $2\sqrt{d-1}$ for any d .

The speed of convergence depends strongly on the base graph: the K_5 family sits within 10^{-3} of the limiting Ramanujan probability from $V = 1000$ onward (0.83258 at $V = 1000$), while the permutation model is still at 0.87282 at $V = 10,000$, and the K_5 ratio g_V is already 0.9522 at $V = 1000$ against the permutation model's 1.148 at ten times that size. The limiting constants are universal in the base graph, however the finite size corrections are not, and the fixed base appears to suppress them. In the same direction, the compensated amplitudes of the K_5 family descend to 1.018 and 1.066 at $V = 10,000$, approaching the regular graph constants 1.0047 and 1.0558 of (3.1), and so the amplitude constant of Conjecture 3 for K_5 appears to be $c(4)$ itself, in the convention of (1.2) with V the cover size. The $K_5 - e$ amplitudes plateau

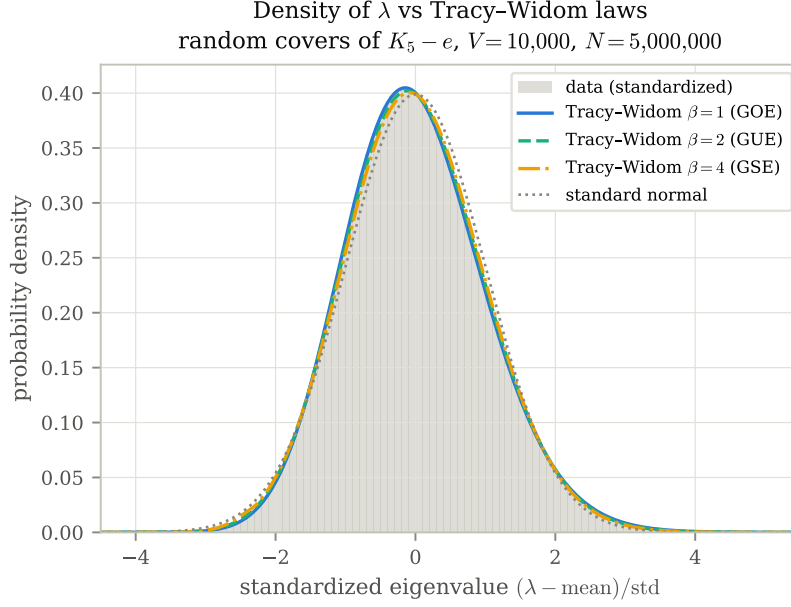


FIGURE 5.2. Standardized empirical density of the largest new eigenvalue over 5×10^6 random covers of $K_5 - e$ at $V = 10,000$, against the standardized Tracy-Widom densities and the standard normal.

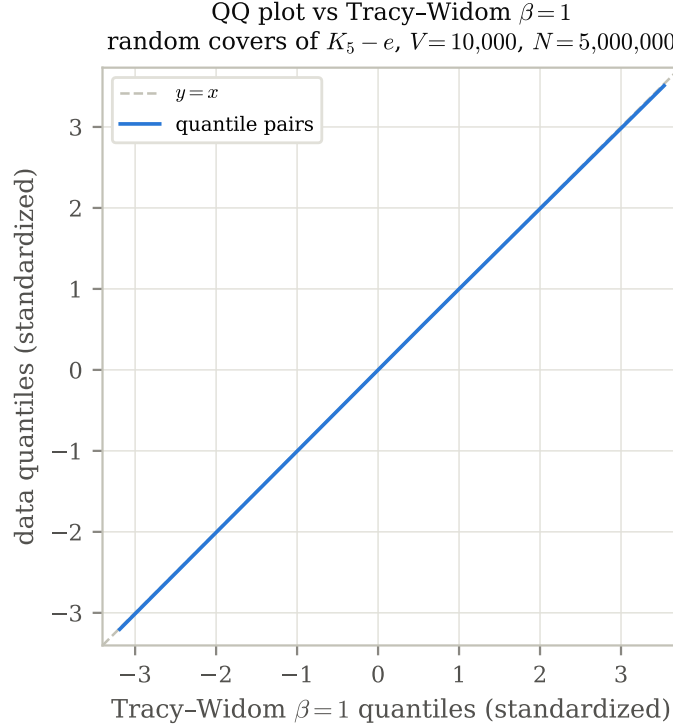


FIGURE 5.3. Quantile-quantile plot of the largest $K_5 - e$ dataset against $\widehat{TW}_1$, probability range 10^{-4} to 0.999: the bulk of the law around the algebraic centering $\rho(K_5 - e)$ is Tracy-Widom at quantile resolution.

lower, with $\sigma_V V^{2/3}$ descending through 0.893 and the compensated gap through 0.839 at $V = 10,000$, and so the $K_5 - e$ analog of $c(d)$ is a new constant, approximately 0.69 against $c(4) = 0.8327$ (the fits $A + BV^{-\theta}$ to the compensated standard deviation and gap give 0.686 and 0.691), the first measurement of an edge scaling constant for an irregular base graph. These amplitudes are stated with V the cover size, and in the $k^{-2/3}$ normalization of Conjecture 3, with $V = 5k$, they multiply by $5^{2/3}$.

5.2. Localized Defects and the Far Tails. The third and fourth moments tell a different story, and the accounting is instructive. The excess kurtosis of the $K_5 - e$ family grows with V , from 0.26 at $V = 100$ to 2.52 at $V = 10,000$ against the TW_1 value 0.165, and its skewness drifts upward through 0.361 against 0.293, while the same statistics for K_5 are 0.28 and 0.292. Figure 5.5 locates the cause in a thin tail of far outliers, at frequencies near 10^{-5} : at $V = 10,000$ a fraction 9×10^{-6} of the $K_5 - e$ samples lie above $\rho + 10\sigma_V$, while the extreme sample of the simple 4-regular family sits only 9 standard deviations out. These outliers are the tangles of [28] made visible sample by sample, and they admit a complete quantitative description, one half of which we can prove.

Note that every subgraph of a cover projects to the base graph by a locally injective map, since distinct neighbors of a vertex of a cover lie over distinct neighbors of its projection. With this in mind, fix a connected graph H of minimum degree at least two admitting a locally injective map, an *immersion*, $\varphi : H \rightarrow B$, and write $r(H) = e(H) - v(H) + 1$ for its cycle rank. Let H^* be the infinite graph obtained from H by attaching, at each vertex x of H and each neighbor of $\varphi(x)$ not used by an edge of H at x , the corresponding branch of the universal cover of B , and define the *defect eigenvalue*

$$\lambda_B(H) = \|A_{H^*}\|,$$

the spectral norm of the adjacency operator of H^* , maximized over the locally injective maps φ when there are several. Deep inside any branch, H^* contains arbitrarily large balls of the universal cover, and so $\lambda_B(H) \geq \rho(B)$ always, and when the inequality is strict we say that H *binds* an eigenvalue above $\rho(B)$. When it binds, $\lambda_B(H)$ lies above the essential spectrum, since removing finitely many vertices from H^* leaves a union of branches, and $\lambda_B(H)$ is computable to any precision from the branch values $y_{u \rightarrow v}(\lambda)$, the fixed point of

$$y_{u \rightarrow v} = \left(\lambda - \sum_{w \sim v, w \neq u} y_{v \rightarrow w} \right)^{-1}$$

over the directed edges of B , which exists for $\lambda > \rho(B)$: each attached branch contributes its value to the diagonal, and $\lambda_B(H)$ is the largest λ with $\lambda_{\max}(A_H + D(\lambda)) = \lambda$, where $D(\lambda)$ is the diagonal matrix of attached branch values (for eigenvalues of graphs with attached infinite tails see also [35]). Two elementary facts organize the far tails: copies of H appear in a random cover at a polynomial rate governed by the cycle rank, and any copy forces a new eigenvalue near $\lambda_B(H)$.

Proposition 5. *Let B be a connected base graph, let H be a connected graph of minimum degree at least two admitting a locally injective map to B , and let X_H be the number of subgraphs of a uniformly random degree k cover H_k of B isomorphic to H . Then*

$$E[X_H] = c_H k^{1-r(H)} (1 + O(k^{-1})), \quad c_H = \frac{J(H, B)}{|\text{Aut}(H)|},$$

where $J(H, B)$ is the number of locally injective homomorphisms from H to B , and, when $r(H) \geq 2$,

$$P(X_H \geq 1) = c_H k^{1-r(H)} (1 + o(1)).$$

Proof. Suppose first that B is simple, as are all of our fixed bases, since loops and multiple edges require only notational changes. Copies of H in the cover correspond to pairs consisting of a locally injective homomorphism $\psi : H \rightarrow B$ and a lift of ψ , with each copy counted $|\text{Aut}(H)|$ times. Fix ψ , and choose for each vertex of H a lift in the fiber over its image, injectively within each fiber, which can be done in $k^{v(H)} (1 + O(k^{-1}))$ ways. The chosen vertices span a copy of H exactly when, for each edge e of B , the permutation σ_e maps every specified source to its specified target. If m_e edges of H lie over e , then the sources are distinct and the targets are distinct, by fiber injectivity and the local injectivity of ψ , and so this event has probability $(k - m_e)!/k! = k^{-m_e} (1 + O(k^{-1}))$, independently across the edges of B . Multiplying and summing over ψ gives the expectation, since $v(H) - e(H) = 1 - r(H)$.

For the probability, we have that $E[X_H^2] = E[X_H] + E[Y]$, where Y counts ordered pairs of distinct copies. The union of two distinct copies is either disconnected, and such pairs contribute $c_H^2 k^{2-2r} (1 + O(k^{-1}))$ by the computation above, or connected of cycle rank at least $r+1$: if two distinct copies overlap, their intersection is a proper subgraph F of H , and $r(U) = 2r(H) - 1 - r(F) + q$ for the union U , where q is the number of components of F and $r(F) = e(F) - v(F) + q$, so it suffices that $r(F) \leq r(H) - 2 + q$. For connected F , with $q = 1$, this says $r(F) \leq r(H) - 1$, which holds since a proper connected subgraph of a graph of minimum degree two has strictly smaller cycle rank: a path leaving the subgraph cannot terminate, and closes an independent cycle whether it returns to the subgraph or to itself. For disconnected F , with $q \geq 2$, it follows from $r(F) \leq r(H)$, since the cycle spaces of disjoint subgraphs embed independently into that of H . Each connected union type of cycle rank at least $r+1$ contributes $O(k^{-r})$, and there are finitely many, and so $E[X_H^2] = E[X_H] (1 + o(1)) + E[X_H]^2 (1 + o(1))$. The claim follows from $P(X_H \geq 1) \geq E[X_H]^2 / E[X_H^2]$ and $P(X_H \geq 1) \leq E[X_H]$. $\square$

The second fact is that every occurrence produces its eigenvalue.

Proposition 6. *In the setting of Proposition 5, suppose that $r(H) \geq 2$ and $\lambda_B(H) > \rho(B)$. Then for every $\epsilon > 0$ there is a constant $c = c(B, H, \epsilon) > 0$ such that for all k sufficiently large,*

$$P(\lambda_{\text{new}}(H_k) \geq \lambda_B(H) - \epsilon) \geq c k^{1-r(H)}.$$

Proof. Let H_ℓ^* denote the truncation of H^* at depth ℓ from H . Since H^* is connected and locally finite, $\lambda_{\max}(H_\ell^*)$ increases to $\|A_{H^*}\|$ as $\ell \rightarrow \infty$, and so we may fix ℓ with $\lambda_{\max}(H_\ell^*) > \lambda_B(H) - \epsilon/2$. Let $u \geq 0$ be the Perron eigenvector of H_ℓ^* with $\|u\| = 1$, supported on $N = N(\ell)$ vertices. Call a copy of H in the cover *clean* if the depth ℓ neighborhood of its vertex set is isomorphic, over B , to H_ℓ^* . A copy which is not clean lies in a connected subgraph of the cover of size $O_\ell(1)$ and cycle rank at least $r(H) + 1$, since any deviation of the neighborhood from H_ℓ^* closes an additional independent cycle within distance ℓ , and the expected number of these is $O_\ell(k^{-r(H)})$ by the computation of Proposition 5, and so with probability at least $c_H k^{1-r(H)} (1 + o(1))$ the cover contains a clean copy.

Given a clean copy, transplant u onto its neighborhood. The subspace of functions which are constant on every fiber is invariant under the adjacency matrix A of the cover, its spectrum on this subspace is the spectrum of B , and its spectrum on the orthogonal complement is exactly the new spectrum. Writing P for the orthogonal projection onto

the fiber constant functions, each fiber meets the support of u in at most N vertices, and so by the Cauchy-Schwarz inequality

$$\|Pu\|^2 = \sum_{b \in V(B)} \frac{1}{k} \left(\sum_{x \in \pi^{-1}(b)} u(x) \right)^2 \leq \frac{N}{k}.$$

Set $w = u - Pu$, a vector in the new subspace with $\|w\|^2 \geq 1 - N/k$. Since $u \geq 0$ and the neighborhood of the clean copy contains H_ℓ^* as a subgraph, we have that $\langle u, Au \rangle \geq \lambda_{\max}(H_\ell^*)$, and since $\|A\|$ is at most the maximum degree of B ,

$$\lambda_{\text{new}}(H_k) \geq \frac{\langle w, Aw \rangle}{\|w\|^2} \geq \langle u, Au \rangle - 3 \|A\| \|Pu\| \geq \lambda_{\max}(H_\ell^*) - O_\ell(k^{-1/2}),$$

whose right hand side is positive and exceeds $\lambda_B(H) - \epsilon$ for k large, and the result follows. $\square$

One family of defects is degenerate. A graph which is itself a cover of B , beginning with B under the identity, saturates every direction at every vertex, and so a copy of it is a full connected component, with defect eigenvalue its own Perron eigenvalue: its occurrence disconnects the cover. Throughout we condition on connected covers, which the pipeline enforces by removing the base spectrum by value (section 2.3), and we exclude covers of B from the defect catalogue and from the minimum in the conjecture below. By Proposition 5 the conditioning changes probabilities by a factor $1 + O(k^{1-r(B)})$. We believe that the remaining defects account for the entire polynomial tail, and conjecture the matching upper bound.

Conjecture 7. *Let B be a connected base graph of cycle rank at least two, and let H_k be a uniformly random degree k cover of B , conditioned on connectedness. Then for every $\epsilon > 0$,*

$$P(\lambda_{\text{new}}(H_k) > \rho(B) + \epsilon) = k^{1-r(\epsilon)+o(1)},$$

where

$$r(\epsilon) = \min \{r(H) : H \text{ immerses in } B \text{ and is not a cover of } B, \lambda_B(H) > \rho(B) + \epsilon\},$$

and where the probability is smaller than every power of k when no such H exists.

For regular Ramanujan bases, Conjecture 7 is essentially the theorem of Friedman and Kohler [32, 31], and the coexistence of a Tracy-Widom bulk with a thin family of localized outliers is the picture proven at the sparse Erdős-Rényi edge [6, 7, 8, 40], and for sparse Wigner matrices the outliers are likewise produced by localized defects [68]. The mechanism is not tied to randomness: Alon, Ganguly, and Srivastava [4] construct regular graphs of high girth whose top eigenvectors are fully localized on small vertex sets, and Alon and Wei [5] prove that every value in $[2\sqrt{d-1}, d]$ is a limit point of second eigenvalues of d -regular graphs. Proposition 6 is the elementary half in general. Together with the fluctuation conjecture, the defect picture determines exactly which statistics converge.

Corollary 8. *Assume the scaling of Conjecture 3, and suppose that some H immersing in B , not a cover of B , has $\lambda_B(H) > \rho(B)$, with $r^* \geq 2$ the least cycle rank of such an H . Then the p^{th} absolute moment of $(\lambda_{\text{new}}(H_k) - \rho(B)) k^{2/3}$ diverges as $k \rightarrow \infty$ for every $p > 3(r^* - 1)/2$.*

To see this, note that a defect at fixed height $\Delta = \lambda_B(H) - \rho(B)$ sits at $\Delta/\sigma_k \asymp \Delta k^{2/3}$ standardized units above the mean, and occurs with probability $\asymp k^{1-r^*}$ by Propositions 5 and 6, and so contributes to the p^{th} standardized moment at order

$$(5.1) \quad k^{1-r^*} (\Delta k^{2/3})^p \asymp k^{1-r^* + \frac{2p}{3}},$$

which diverges exactly when $p > 3(r^* - 1)/2$. Under Conjecture 7, holding uniformly down to the fluctuation scale, the moments below the threshold converge instead to those of $c_B TW_1$. Equation (5.1) organizes every moment anomaly in this paper. For the permutation multigraph model, whose loops and multiple edges are binding defects of cycle rank one, even the tail mass is of constant order and the law itself fails, which is what section 3.4 observed. For $K_4 - e$, where $r^* = 2$, the variance itself is eventually dominated by the defect band, growing relative to the bulk like $k^{1/3}$, and at the bulk scale the skewness diverges like k and the kurtosis like $k^{5/3}$, so that the raw moments of this family are the most contaminated in the dataset, and the raw ratio g_V of section 5.3 must eventually turn around and decrease, with only its trimmed version converging to the TW_1 ratio. For $K_5 - e$ and K_5 , where $r^* = 3$, the variance is safe, the kurtosis diverges like $k^{2/3}$, matching the observed growth, and $p = 3$ is exactly critical: the defects add a constant, non-vanishing offset to the skewness, which is why the $K_5 - e$ skewness settles near 0.36 rather than at the Tracy-Widom value 0.293, while for K_5 the constant is too small to resolve. Every quantile level statistic is immune, since the total defect mass vanishes: the Kolmogorov-Smirnov distance moves by at most the tail mass and the Ramanujan probability by less, and this is why the fixed base families are omitted from Figure 4.4 while every distributional test of section 5.1 converges.

The data confirms the computed defects in detail. Evaluating $\lambda_B(H)$ by the branch recursion, and confirming the values by planting the subgraphs in sampled covers, we have that for $K_5 - e$ no subgraph of cycle rank two binds an eigenvalue above ρ , that K_4 , of cycle rank three, binds

$$\lambda_{K_5-e}(K_4) = 3.3391340785 \dots,$$

with its subdivisions filling the interval below it, and that the largest cycle rank four value is $3.4519402 \dots$, while a disjoint copy of $K_5 - e$ itself would give $1 + \sqrt{7}$. The outlier values at the largest sizes fill an interval above ρ , leave a gap, and cluster on the K_4 atom, where the extreme samples at $V = 5000$ and $10,000$ sit, and the extremes at $V \leq 2000$ sit instead on the rank four atom, which decays faster and is gone by $V = 5000$ (Figure 5.4). The frequencies obey Proposition 5: the count in a window around the K_4 atom decays with fitted exponent 2.02 ± 0.13 against the predicted k^{-2} , and at thresholds beyond the reach of cycle rank three the fitted exponent is 2.90 ± 0.14 against k^{-3} . The apparent lengthening of the tail, from 27 standard deviations at $V = 5000$ to 42 at $V = 10,000$, is then exact: a fixed atom measured in units of $\sigma_V \propto V^{-2/3}$ lengthens by $2^{2/3} = 1.587$, and the observed ratio is 1.584. For the K_5 family the same computation gives the atom

$$\lambda_{K_5}(K_4) = \frac{7}{2}$$

exactly, and the six samples more than nine standard deviations above the mean at $V = 10,000$ all lie in $[3.497, 3.510]$. Table 2 collects the catalogue across all three fixed bases, with the $K_4 - e$ values derived in subsection 5.3.

Far outliers lock onto the localized defect eigenvalues (dotted); frequencies decay like k^{1-r}

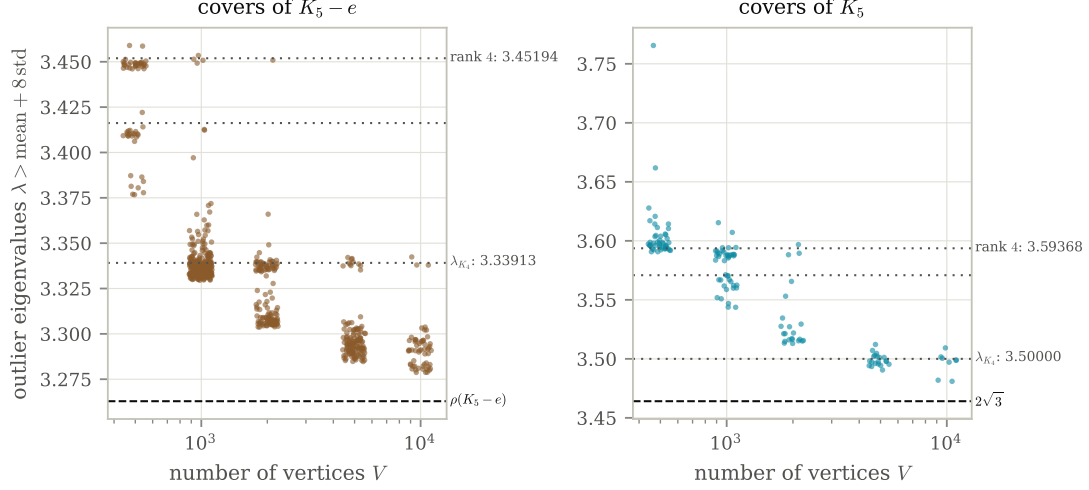


FIGURE 5.4. Every sample more than eight standard deviations above the mean, against V , for the $K_5 - e$ (left) and K_5 (right) cover families, with the computed defect eigenvalues overlaid: the outliers lock onto the K_4 atoms, $3.33913\dots$ for $K_5 - e$ and exactly $7/2$ for K_5 , and onto the cycle rank four values (dotted), while the continuum of K_4 subdivisions drains toward ρ and the frequencies decay like k^{1-r} .

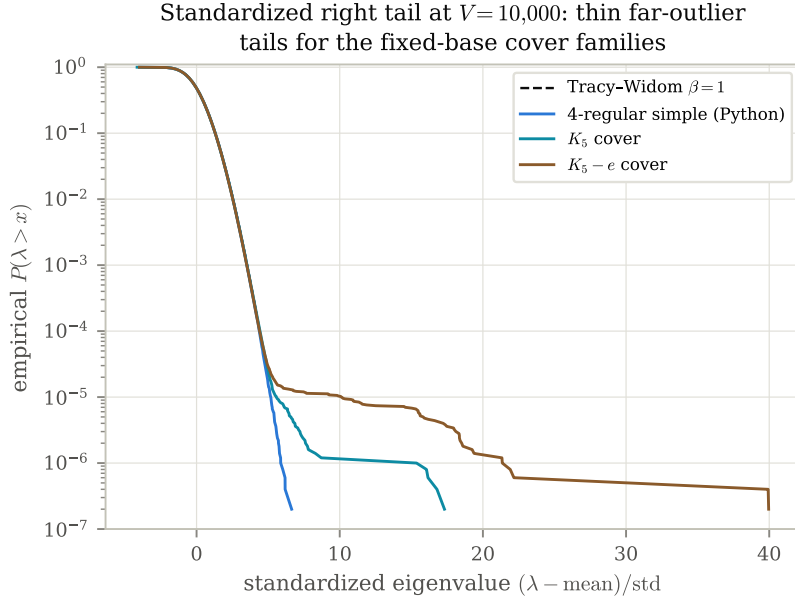


FIGURE 5.5. Standardized right tail $P(\lambda > x)$ at $V = 10,000$ for the simple 4-regular, K_5 cover, and $K_5 - e$ cover families. The bulk of all three collapses onto the Tracy-Widom $\beta = 1$ tail, while the $K_5 - e$ family carries a thin far outlier tail at frequency near 10^{-5} , reaching 42 standard deviations, and the K_5 family a much thinner one.

base B	defect H	$r(H)$	$\lambda_B(H)$	rate k^{1-r}	fitted exponent
$K_5 - e$	K_4	3	3.3391340785...	k^{-2}	2.02 ± 0.13
$K_5 - e$	largest rank four	4	3.4519402...	k^{-3}	2.90 ± 0.14
K_5	K_4	3	7/2	k^{-2}	1.7 ± 0.7
$K_4 - e$	$\theta(2, 2, 4)$	2	2.5118521...	k^{-1}	≈ 1.3 , descending
$K_4 - e$	$\theta(1, 2, 5)$	2	2.5136918...	k^{-1}	≈ 1.3 , descending
$K_4 - e$	rank three band	3	$\lesssim 2.530$	k^{-2}	1.96

TABLE 2. The catalogue of leading localized defects for the three fixed base families. Each row lists a subgraph H immersing in B , its cycle rank $r(H)$, the defect eigenvalue $\lambda_B(H)$ computed by the branch recursion, the occurrence rate $k^{1-r(H)}$ of Proposition 5, and the exponent fitted to the observed outlier counts. Here $\theta(a, b, c)$ is the generalized theta graph whose two branch vertices are joined by routes of lengths a , b , and c , and the two $K_4 - e$ atoms sit too close together for separate counts, so one exponent is fitted to the pair.

5.3. The Boundary Case $K_4 - e$. The third fixed base, $K_4 - e$, tests Conjecture 3 at its boundary: the complete graph on four vertices minus an edge has cycle rank exactly two, degrees $(2, 3, 3, 2)$, and the closed form

$$\rho(K_4 - e) = \sqrt{1 + 2\sqrt{7}} = 2.5082867902\dots$$

Suppressing its two degree two vertices identifies $K_4 - e$ with the theta graph, two vertices joined by three parallel edges, and so random covers of $K_4 - e$ are subdivisions of random theta covers, which are exactly the bipartite cubic multigraphs built from three independent uniform permutations, the bipartite model of MNS. We sampled 10^6 covers per size at the same seven sizes $V = 4k$ between 100 and 10,000.

The family fits the framework, and it converges more slowly than any other. The mean approaches ρ from below with the ratio g_V rising through 0.7326 at $V = 10,000$ and the Ramanujan probability through 0.79490, both far from their conjectured limits $|\mu_{TW_1}|/\sigma_{TW_1}$ and $F_1(0)$. The raw distributional statistics are contaminated in a way the other families only hint at: the excess kurtosis grows to 12.7, and the Kolmogorov-Smirnov distance to $\widetilde{TW}_1$, after falling to 3.8×10^{-3} at $V = 500$, grows again to 1.9×10^{-2} at $V = 10,000$ (Figure 5.6, left). The cause is the far tail, and removing it, by discarding the 0.2% of samples beyond six standard deviations, restores $\widetilde{TW}_1$ as the nearest law from $V = 1000$ on, with distance 4.0×10^{-3} at $V = 10,000$ against 6.4×10^{-3} and 9.9×10^{-3} for $\widetilde{TW}_2$ and $\widetilde{TW}_4$.

The tail itself differs from the $K_5 - e$ tail exactly as the defect picture predicts. No cycle rank one subgraph binds an eigenvalue above ρ , and no connected 2-cover of $K_4 - e$ has a new eigenvalue above ρ either, as a direct check of all 32 signings shows, however the base now has cycle rank two, and so proper rank two subgraphs exist, generalized thetas whose routes are longer than the base's and which carry branches, occurring at the rate k^{-1} of Proposition 5: the two shortest, $\theta(2, 2, 4)$ and $\theta(1, 2, 5)$ in the notation of Table 2, bind at

$$2.5118521\dots \quad \text{and} \quad 2.5136918\dots,$$

computed by the same branch recursion, one power of k more frequent than the leading $K_5 - e$ defects. Above the rank two band, rank three shapes reach ≈ 2.530 at rate k^{-2} . The data shows precisely this structure (Figure 5.6, right): the outliers form a dense band draining toward ρ rather than isolated atoms, capped near 2.530 at the largest

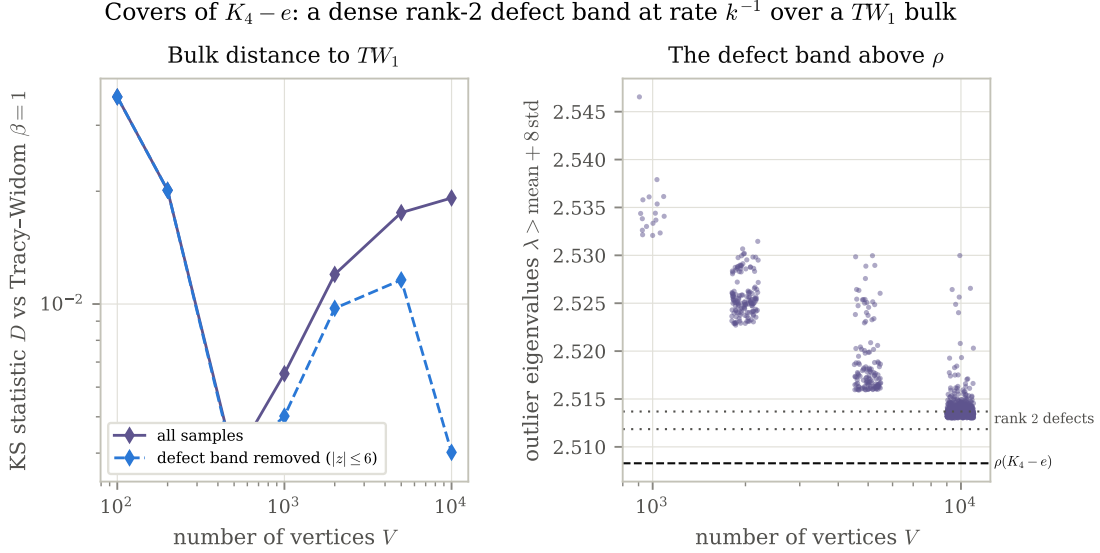


FIGURE 5.6. Covers of $K_4 - e$. Left: Kolmogorov-Smirnov distance of the standardized sample to $\widehat{TW}_1$, for all samples and with the defect band removed. Right: every sample more than eight standard deviations above the mean against V , with $\rho(K_4 - e)$ dashed and the two computed rank two defect eigenvalues dotted. The band drains toward ρ at rate k^{-1} , with rank three shapes reaching ≈ 2.530 at rate k^{-2} .

sizes, with the count above the rank two cap decaying with fitted exponent 1.96 and the count at the cap decaying visibly more slowly, still descending toward the predicted 1. In the notation of Conjecture 7, $r(\epsilon) = 2$ for ϵ below ≈ 0.006 and 3 up to ≈ 0.022 . The comparison across the three fixed bases is the point: the denser the base, the higher the cycle rank at which the defect hierarchy starts, and the further the atoms sit from ρ . For $K_4 - e$ the hierarchy starts at rank two, the band crowds the bulk at accessible sizes, and the moments and even the raw Kolmogorov-Smirnov distance feel it, which is why this family, at the boundary of the conjecture, converges last.

6. OPEN PROBLEMS

The Block Laws. Conjecture 1 is open in every case, including $\beta = 1$: for covers of a d -regular base of size $n \geq 2$ no analog of Theorem 4 is known, and for irregular bases even the correct formulation of edge rigidity around $\rho(B)$ is open. The abelian case seems the natural first target, since the block A_χ is an explicit random Hermitian matrix built from a random regular graph and independent uniform phases.

Problem 9. Prove that the largest eigenvalue of the $\mathbb{Z}_3$ -character block A_χ of a random $\mathbb{Z}_3$ -cover of a random simple d -regular graph has TW_2 fluctuations, in the sense of Conjecture 1.

The Maximum Laws and the Amplitudes. The mixed families raise a cleaner question than convergence alone: the maximum law depends on the relative scale of the blocks, which we conjectured equal for character blocks in section 4.3.

Problem 10. Determine the scale constants c_ρ of Conjecture 1 and c_B of Conjecture 3. In particular, prove that $c_\chi = c(d)$ for every non-trivial character χ of an abelian

cover, determine the quaternionic constant, for which the data suggests $c_\rho = c(d)2^{-1/6}$, and determine c_{K_5-e} , for which the data gives approximately 0.69.

I believe that the equal amplitude prediction $c_\chi = c(d)$ is exactly right, as the one percent discrepancies at $n = 5000$ shrink monotonically with n in our data.

Far Outliers. The permutation multigraph model of section 3.4 converges to the same threshold with the same $V^{-2/3}$ mean gap, however its fluctuations are dominated by localized outliers, and the fixed base families of section 5 carry a much thinner tail of the same character.

Problem 11. Determine the limiting distribution of the largest non-trivial eigenvalue of the permutation multigraph model. Is it a Tracy-Widom law conditioned on the absence of localized eigenvalues, plus an independent point process of outliers?

Problem 12. Prove the upper bound in Conjecture 7: that nothing but the defects of section 5.2 inhabits the polynomial tail. The lower bound is Proposition 6, the regular Ramanujan case is the theorem of Friedman and Kohler [32, 31], and the fitted exponents of sections 5.2 and 5.3 match the conjectured rates on two irregular bases with different leading ranks.

Growing Covers. Huang and Yau [45] proved Tracy-Widom fluctuations for regular graphs of growing degree. The analogous regime for covers, base size fixed and cover group growing, is precisely the setting of Conjecture 3, however one may also grow the base: for $\mathbb{Z}_k$ covers with k growing with n , the number of GUE blocks grows, and the largest new eigenvalue becomes the maximum of a growing number of Tracy-Widom variables, suggesting a transition from Tracy-Widom to Gumbel fluctuations. For iterated 2-covers of dense random regular graphs, Liu and Zou [50] prove exactly this maximum of independent Tracy-Widom structure.

Problem 13. For random $\mathbb{Z}_k$ covers of a random d -regular graph on n vertices, determine the joint regimes of (k, n) in which the largest new eigenvalue has Tracy-Widom, and Gumbel, fluctuations.

Acknowledgements

I would like to thank Lior Silberman, jointly with whom the conjectures of this paper were formulated in 2018, for years of discussions about this project. I would also like to thank Peter Sarnak for suggesting the problem of the distribution of new eigenvalues of random covers, and Stefan Glock for helpful correspondence regarding Sodin's work and the scaling constants.

REFERENCES

- [1] Naman Agarwal, Karthekeyan Chandrasekaran, Alexandra Kolla, and Vivek Madan. On the expansion of group-based lifts. *SIAM Journal on Discrete Mathematics*, 33(3):1338–1373, 2019.
- [2] Noga Alon. Eigenvalues and expanders. *Combinatorica*, 6(2):83–96, 1986.
- [3] Noga Alon. Explicit expanders of every degree and size. *Combinatorica*, 41(4):447–463, 2021.
- [4] Noga Alon, Shirshendu Ganguly, and Nikhil Srivastava. High-girth near-Ramanujan graphs with localized eigenvectors. *Israel Journal of Mathematics*, 246:1–20, 2021.
- [5] Noga Alon and Fan Wei. The limit points of the top and bottom eigenvalues of regular graphs, 2023. Preprint.
- [6] Johannes Alt, Raphaël Ducatez, and Antti Knowles. Extremal eigenvalues of critical Erdős-Rényi graphs. *Annals of Probability*, 49(3):1347–1401, 2021.

- [7] Johannes Alt, Raphaël Ducatez, and Antti Knowles. Poisson statistics and localization at the spectral edge of sparse Erdős–Rényi graphs. *Annals of Probability*, 51(1):277–358, 2023.
- [8] Johannes Alt, Raphaël Ducatez, and Antti Knowles. Localized phase for the Erdős–Rényi graph. *Communications in Mathematical Physics*, 405:Paper No. 9, 2024.
- [9] Alon Amit and Nathan Linial. Random graph coverings I: General theory and graph connectivity. *Combinatorica*, 22(1):1–18, 2002.
- [10] Alon Amit and Nathan Linial. Random lifts of graphs: edge expansion. *Combinatorics, Probability and Computing*, 15(3):317–332, 2006.
- [11] Omer Angel, Joel Friedman, and Shlomo Hoory. The non-backtracking spectrum of the universal cover of a graph. *Transactions of the American Mathematical Society*, 367(6):4287–4318, 2015.
- [12] Roland Bauerschmidt, Jiaoyang Huang, Antti Knowles, and Horng-Tzer Yau. Edge rigidity and universality of random regular graphs of intermediate degree. *Geometric and Functional Analysis*, 30(3):693–769, 2020.
- [13] Yonatan Bilu and Nathan Linial. Lifts, discrepancy and nearly optimal spectral gap. *Combinatorica*, 26(5):495–519, 2006.
- [14] Charles Bordenave. A new proof of Friedman’s second eigenvalue theorem and its extension to random lifts. *Annales Scientifiques de l’École Normale Supérieure (4)*, 53(6):1393–1439, 2020.
- [15] Charles Bordenave and Benoît Collins. Eigenvalues of random lifts and polynomials of random permutation matrices. *Annals of Mathematics*, 190(3):811–875, 2019.
- [16] Charles Bordenave and Benoît Collins. Norm of matrix-valued polynomials in random unitaries and permutations, 2023. Preprint.
- [17] Folkmar Bornemann. On the numerical evaluation of distributions in random matrix theory: a review. *Markov Processes and Related Fields*, 16:803–866, 2010.
- [18] Folkmar Bornemann. On the numerical evaluation of Fredholm determinants. *Mathematics of Computation*, 79(270):871–915, 2010.
- [19] Teodor Bucht, Kevin Schnelli, and Yuanyuan Xu. Quantitative Tracy–Widom laws for sparse random matrices, 2025. Preprint.
- [20] Karthekeyan Chandrasekaran and Ameya Velingker. Shift lifts preserving Ramanujan property. *Linear Algebra and its Applications*, 529:199–214, 2017.
- [21] Chi-Fang Chen, Jorge Garza-Vargas, Joel A. Tropp, and Ramon van Handel. A new approach to strong convergence. *Annals of Mathematics*, 203(2):555–602, 2026.
- [22] Leonard N. Choup. Edgeworth expansion of the largest eigenvalue distribution function of GOE. *Journal of Mathematical Physics*, 50(1):013512, 2009.
- [23] Matthew B. Crawford, David J. Marchette, William Maxwell, and Samuel S. Mendelson. Spectral properties of random graphs with fixed equitable partition, 2023.
- [24] Freeman J. Dyson. The threefold way. Algebraic structure of symmetry groups and ensembles in quantum mechanics. *Journal of Mathematical Physics*, 3:1199–1215, 1962.
- [25] Joel Friedman. On the second eigenvalue and random walks in random d -regular graphs. *Combinatorica*, 11(4):331–362, 1991.
- [26] Joel Friedman. Some geometric aspects of graphs and their eigenfunctions. *Duke Mathematical Journal*, 69(3):487–525, 1993.
- [27] Joel Friedman. Relative expanders or weakly relatively Ramanujan graphs. *Duke Mathematical Journal*, 118(1):19–35, 2003.
- [28] Joel Friedman. A proof of Alon’s second eigenvalue conjecture and related problems. *Memoirs of the American Mathematical Society*, 195(910), 2008.
- [29] Joel Friedman. Formal zeta function expansions and the frequency of Ramanujan graphs. In *Probabilistic Methods in Geometry, Topology and Spectral Theory*, volume 739 of *Contemporary Mathematics*, pages 45–62. American Mathematical Society, 2019.
- [30] Joel Friedman, Jeff Kahn, and Endre Szemerédi. On the second eigenvalue in random regular graphs. In *Proceedings of the 21st Annual ACM Symposium on Theory of Computing (STOC 1989)*, pages 587–598. ACM, 1989.
- [31] Joel Friedman and David Kohler. On the relativized Alon second eigenvalue conjecture I: main theorems, examples, and outline of proof, 2019. Preprint.
- [32] Joel Friedman and David-Emmanuel Kohler. The relativized second eigenvalue conjecture of Alon, 2014. Reorganized as the series “On the Relativized Alon Second Eigenvalue Conjecture I–VI”, arXiv:1911.05688, 1911.05705, 1911.05710, 1911.05721, 1911.06620, 1911.05775 (2019).

- [33] Joel Friedman and Jean-Pierre Tillich. Generalized Alon–Boppana theorems and error-correcting codes. *SIAM Journal on Discrete Mathematics*, 19(3):700–718, 2005.
- [34] Jorge Garza-Vargas and Archit Kulkarni. Spectra of infinite graphs via freeness with amalgamation. *Canadian Journal of Mathematics*, 75(5):1633–1684, 2023.
- [35] Leonid Golinskii. Spectra of infinite graphs with tails. *Linear and Multilinear Algebra*, 64(11):2270–2296, 2016.
- [36] Catherine Greenhill, Svante Janson, Jeong Han Kim, and Nicholas C. Wormald. Permutation pseudographs and contiguity. *Combinatorics, Probability and Computing*, 11(3):273–298, 2002.
- [37] Chris Hall, Doron Puder, and William F. Sawin. Ramanujan coverings of graphs. *Advances in Mathematics*, 323:367–410, 2018.
- [38] Yukun He. Spectral gap and edge universality of dense random regular graphs. *Communications in Mathematical Physics*, 405(8):Paper No. 181, 2024.
- [39] Will Hide and Michael Magee. Near optimal spectral gaps for hyperbolic surfaces. *Annals of Mathematics*, 198(2):791–824, 2023.
- [40] Ella Hiesmayr and Theo McKenzie. The spectral edge of constant degree Erdős–Rényi graphs. *Random Structures & Algorithms*, 66(3):e70011, 2025.
- [41] Shlomo Hoory. A lower bound on the spectral radius of the universal cover of a graph. *Journal of Combinatorial Theory, Series B*, 93(1):33–43, 2005.
- [42] Shlomo Hoory, Nathan Linial, and Avi Wigderson. Expander graphs and their applications. *Bulletin of the American Mathematical Society*, 43(4):439–561, 2006.
- [43] Jiaoyang Huang, Theo McKenzie, and Horng-Tzer Yau. Optimal eigenvalue rigidity of random regular graphs, 2024. Preprint.
- [44] Jiaoyang Huang, Theo McKenzie, and Horng-Tzer Yau. Ramanujan property and edge universality of random regular graphs, 2024.
- [45] Jiaoyang Huang and Horng-Tzer Yau. Edge universality of random regular graphs of growing degrees, 2023.
- [46] Jiaoyang Huang and Horng-Tzer Yau. Spectrum of random d -regular graphs up to the edge. *Communications on Pure and Applied Mathematics*, 77(3):1635–1723, 2024.
- [47] Dmitry Jakobson, Stephen D. Miller, Igor Rivin, and Zeév Rudnick. Eigenvalue spacings for regular graphs. In *Emerging Applications of Number Theory (Minneapolis, MN, 1996)*, volume 109 of *The IMA Volumes in Mathematics and its Applications*, pages 317–327. Springer, New York, 1999.
- [48] Iain M. Johnstone and Zongming Ma. Fast approach to the Tracy–Widom law at the edge of GOE and GUE. *Annals of Applied Probability*, 22(5):1962–1988, 2012.
- [49] Nati Linial and Doron Puder. Word maps and spectra of random graph lifts. *Random Structures & Algorithms*, 37(1):100–135, 2010.
- [50] Dang-Zheng Liu and Guangyi Zou. Edge universality for inhomogeneous random matrices, 2025.
- [51] Eyal Lubetzky, Benny Sudakov, and Van Vu. Spectra of lifted Ramanujan graphs. *Advances in Mathematics*, 227(4):1612–1645, 2011.
- [52] A. Lubotzky, R. Phillips, and P. Sarnak. Ramanujan graphs. *Combinatorica*, 8(3):261–277, 1988.
- [53] Michael Magee, Frédéric Naud, and Doron Puder. A random cover of a compact hyperbolic surface has relative spectral gap $\frac{3}{16} - \epsilon$. *Geometric and Functional Analysis*, 32:595–661, 2022.
- [54] Michael Magee, Doron Puder, and Ramon van Handel. Strong convergence of uniformly random permutation representations of surface groups, 2025. Preprint.
- [55] Yotam Maoz. Smooth linear eigenvalue statistics on random covers of compact hyperbolic surfaces—a central limit theorem and almost sure RMT statistics. *Israel Journal of Mathematics*, 2026. Published online.
- [56] Adam W. Marcus, Daniel A. Spielman, and Nikhil Srivastava. Interlacing families I: Bipartite Ramanujan graphs of all degrees. *Annals of Mathematics*, 182(1):307–325, 2015.
- [57] G. A. Margulis. Explicit group-theoretic constructions of combinatorial schemes and their applications in the construction of expanders and concentrators. *Problemy Peredachi Informatsii*, 24(1):51–60, 1988.
- [58] Brendan McKay. Answer to “Can we calculate the spectral radius of the universal cover for specific graphs?”. MathOverflow, 2023. <https://mathoverflow.net/a/440155> (version: 2023-11-19).
- [59] Steven J. Miller, Tim Novikoff, and Anthony Sabelli. The distribution of the largest nontrivial eigenvalues in families of random regular graphs. *Experimental Mathematics*, 17(2):231–244, 2008.

- [60] Sidhanth Mohanty and Ryan O'Donnell. *X-Ramanujan graphs*. In *Proceedings of the 2020 ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pages 1226–1243, 2020.
- [61] Julien Moy. Spectral statistics of the Laplacian on random covers of a closed negatively curved surface, 2024. Preprint.
- [62] A. Nilli. On the second eigenvalue of a graph. *Discrete Mathematics*, 91(2):207–210, 1991.
- [63] A. Nilli. Tight estimates for eigenvalues of regular graphs. *Electronic Journal of Combinatorics*, 11(1):Note N9, 2004.
- [64] Doron Puder. Expansion of random graphs: new proofs, new results. *Inventiones Mathematicae*, 201(3):845–908, 2015.
- [65] Peter Sarnak. What is... an expander? *Notices of the American Mathematical Society*, 51(7):762–763, 2004.
- [66] Kevin Schnelli and Yuanyuan Xu. Convergence rate to the Tracy–Widom laws for the largest eigenvalue of Wigner matrices. *Communications in Mathematical Physics*, 393(2):839–907, 2022.
- [67] Sasha Sodin. The Tracy–Widom law for some sparse random matrices, 2009. arXiv:0903.4295. Published as J. Stat. Phys. 136 (2009), no. 5, 834–841, doi:10.1007/s10955-009-9813-2; see also the Erratum, J. Stat. Phys. 166 (2017), no. 5, 1343, doi:10.1007/s10955-017-1715-0. The arXiv posting was withdrawn (v6, 18 December 2016) with the comment: “The proof contains a serious mistake, and Lemma 5 can not be true as stated. For example, the probability that the graph is not connected tends to zero slower than eq. 4 would imply. I am grateful to Charles Bordenave and Sandrine Péché who brought this to my attention”.
- [68] Konstantin Tikhomirov and Pierre Youssef. Outliers in spectrum of sparse Wigner matrices. *Random Structures & Algorithms*, 58(3):517–605, 2021.
- [69] Craig A. Tracy and Harold Widom. Level-spacing distributions and the Airy kernel. *Communications in Mathematical Physics*, 159(1):151–174, 1994.
- [70] Craig A. Tracy and Harold Widom. On orthogonal and symplectic matrix ensembles. *Communications in Mathematical Physics*, 177(3):727–754, 1996.
- [71] Ramon van Handel. The strong convergence phenomenon, 2025. Preprint.
- [72] Nicholas C. Wormald. Models of random regular graphs. In *Surveys in Combinatorics, 1999 (Canterbury)*, volume 267 of *London Mathematical Society Lecture Note Series*, pages 239–298. Cambridge University Press, Cambridge, 1999.

Email address: naslund.math@gmail.com